

Part I

Feynman Path Integral

Chapter 1

Feynman Path Integral

Two formulations of quantum mechanics:

1. Schrödinger's wave function (operator form):

$$\hat{H}|\psi\rangle = i\hbar \frac{\partial}{\partial t}|\psi\rangle \quad (1.1)$$

2. Feynman's path integral (c-number form):

$$iG(\text{Green's function}) \propto \int \mathcal{D}(x, t) e^{i \int \mathcal{L} dt} \quad (1.2)$$

There are many advantages of the Feynman path integral:

1. It makes the double-slit experiment more understandable.
2. The classical limit “ $\hbar \rightarrow 0$ ” is “tractable”: quantum $\xrightarrow{\hbar \rightarrow 0}$ classical.
3. It provides a semiclassical picture of quantum mechanics.
4. “Quantum fluctuations” are more “understandable”.
5. A natural route to the low-energy effective theory of quantum many-body systems.
6. A natural language for describing topological properties of quantum many-body systems.

However, practical calculations in the path-integral representation of simple quantum-mechanical problems may be notoriously difficult and lengthy.

1.1 Propagators

Consider a quantum particle confined in a one-dimensional space:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \quad (1.3)$$

and the canonical pair: $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$

The Schrödinger equation is:

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H}|\psi(t)\rangle \quad (1.4)$$

This first-order nature allows us to define a time evolution operator $\hat{U}(t, t_0)$ which propagates the state vector from an initial time t_0 to a final time t :

$$|\psi(t)\rangle = \hat{U}(t, t_0)|\psi(t_0)\rangle \quad (1.5)$$

Assuming the Hamiltonian H does not explicitly depend on time, the formal solution of $\hat{U}$ is:

$$\hat{U} = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)} \quad (1.6)$$

A crucial property of $\hat{U}$ is the “chain-like” rule, or composition property. For any intermediate time t' such that $t > t' > t_0$:

$$\hat{U}(t, t_0) = \hat{U}(t, t')\hat{U}(t', t_0) \quad (1.7)$$

This property is the key to the entire path integral derivation. Moreover, $\hat{U}(t, t_0)$ is unitary:

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{\mathbb{I}} \quad (1.8)$$

where $\hat{\mathbb{I}}$ is the identity operator.

In the position representation, we can obtain matrix elements:

$$\begin{aligned} U(x, t; x_0, t_0) &= \langle x | \hat{U}(t, t_0) | x_0 \rangle \\ &= \langle x | e^{-\frac{i}{\hbar}\hat{H}(t-t_0)} | x_0 \rangle \end{aligned} \quad (1.9)$$

We can define a propagator (Green’s function) of the quantum system using these matrix elements:

$$iG(x, t; x_0, t_0) = U(x, t; x_0, t_0) \quad (1.10)$$

Using the matrix elements, $\psi(x, t)$ can be rewritten as:

$$\begin{aligned} \psi(x, t) &= \langle x | \hat{U}(t, t_0) | \psi(t_0) \rangle \\ &= \int dx_0 \langle x | \hat{U}(t, t_0) | x_0 \rangle \langle x_0 | \psi(t_0) \rangle \\ &= \int dx_0 U(x, t; x_0, t_0) \psi(x_0, t_0) \end{aligned} \quad (1.11)$$

The propagator also satisfies the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} G(x, t; x_0, t_0) = \hat{H} G(x, t; x_0, t_0) \quad (1.12)$$

The initial condition is:

$$G(x, t_0; x_0, t_0) = -i \langle x | \hat{U}(t_0, t_0) | x_0 \rangle = -i \delta(x - x_0) \quad (1.13)$$

$\delta(x - x_0)$ is the Dirac delta function.

Example 1.1.1. For free particle: $\hat{H} = \frac{1}{2m} \hat{p}^2$, in position representation:

$$\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x} \quad (1.14)$$

So the PDE is:

$$i\hbar \frac{\partial}{\partial t} G(x, t; x_0, t_0) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} G(x, t; x_0, t_0) \quad (1.15)$$

Solving the PDE:

Using the Fourier transform: (we use $G(x, t)$ instead of $G(x, t; x_0, t_0)$). We solve the free-particle Green's function by transforming to momentum space:

$$\begin{cases} G(x, t) = \frac{1}{\sqrt{2\pi}} \int dk \cdot \tilde{G}(k, t) e^{ikx} \\ \tilde{G}(k, t) = \frac{1}{\sqrt{2\pi}} \int dx G(x, t) e^{-ikx} \end{cases} \quad (1.16)$$

With these conventions, spatial derivatives become algebraic in k -space while the time derivative remains unchanged:

$$\begin{cases} \mathcal{F} \left\{ i\hbar \frac{\partial G(x, t)}{\partial t} \right\} = i\hbar \frac{\partial \tilde{G}(k, t)}{\partial t} \\ \mathcal{F} \left\{ -\frac{\hbar^2}{2m} \frac{\partial^2 G}{\partial x^2} \right\} = -\frac{\hbar^2}{2m} [-k^2 \tilde{G}(k, t)] = \frac{\hbar^2 k^2}{2m} \tilde{G}(k, t) \end{cases} \quad (1.17)$$

Applying the transform to the PDE yields an ordinary differential equation in time for each k :

$$\begin{cases} i\hbar \frac{\partial}{\partial t} \tilde{G}(k, t) = \frac{\hbar^2 k^2}{2m} \tilde{G}(k, t) \\ \frac{d\tilde{G}}{\tilde{G}} = -i \frac{\hbar k^2}{2m} dt \end{cases} \quad (1.18)$$

Integrating in time gives the logarithm of the solution up to a k -dependent constant:

$$\ln \tilde{G} = -i \frac{\hbar k^2}{2m} t + C(k) \quad (1.19)$$

So we can get the solution:

$$\tilde{G}(k, t) = A(k) e^{-i \frac{\hbar k^2}{2m} t}, \quad A(k) = e^{C(k)} \quad (1.20)$$

To determine $A(k)$, impose the initial condition at time t_0 in position space:

$$\begin{aligned}\tilde{G}(k, t_0) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx G(x, t_0) e^{-ikx} \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \delta(x - x_0) e^{-ikx}\end{aligned}\quad (1.21)$$

Using the Fourier transform of the Dirac delta, we find:

$$\tilde{G}(k, t_0) = \frac{1}{\sqrt{2\pi}} e^{-ikx_0} \quad (1.22)$$

Matching at t_0 fixes the k -space amplitude:

$$A(k) = \frac{1}{\sqrt{2\pi}} e^{-ikx_0} \cdot e^{i\frac{\hbar k^2}{2m} t_0}. \quad (1.23)$$

Therefore, for general time t we have:

$$\tilde{G}(k, t) = \frac{1}{\sqrt{2\pi}} e^{-ikx_0} e^{-i\frac{\hbar k^2}{2m}(t-t_0)} \quad (1.24)$$

Finally, inverse-transform back to position space to obtain the integral representation of the propagator:

$$\begin{aligned}G(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \cdot \tilde{G}(k, t) e^{ikx} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ik(x-x_0)} \cdot e^{-i\frac{\hbar(t-t_0)}{2m} k^2}\end{aligned}\quad (1.25)$$

This is a standard Gaussian integral of the form:

$$\int_{-\infty}^{\infty} dk e^{-ak^2+bk} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}. \quad (1.26)$$

Let's identify the coefficients:

$$\begin{cases} a = i\frac{\hbar(t-t_0)}{2m} \\ b = i(x-x_0) \end{cases} \quad (1.27)$$

So we can get the solution:

$$iG(x, t) = \left[\frac{m}{2\pi\hbar i(t-t_0)} \right]^{\frac{1}{2}} \cdot e^{i\frac{\hbar}{2m} \frac{m(x-x_0)^2}{2(t-t_0)}} \quad (1.28)$$

Alternatively, we can solve this PDE directly from the definition:

$$\begin{aligned}iG &= \langle x | e^{-\frac{i}{\hbar}(t-t_0)\hat{H}} | x_0 \rangle \\ &= \int \langle x | e^{-\frac{i}{\hbar}(t-t_0)\hat{H}} | p \rangle \langle p | x_0 \rangle dp \\ &= \int dp e^{-\frac{i(t-t_0)p^2}{2m\hbar}} \langle x | p \rangle \langle p | x_0 \rangle \\ &= \frac{1}{2\pi\hbar} \int dp e^{-\frac{i(t-t_0)p^2}{2m\hbar} + i\frac{(x-x_0)}{\hbar} p}.\end{aligned}\quad (1.29)$$

Using $P = \hbar k$, we obtain the same result as in the Fourier-transform method.

1.2 Path-Integral

When $t > t_1 > t_0$ and t_1 is an arbitrarily chosen intermediate time, we can write:

$$\begin{aligned}
 iG(x, t; x_0, t_0) &= \langle x | \hat{U}(t, t_0) | x_0 \rangle \\
 &= \langle x | \hat{U}(t, t_1) \hat{U}(t_1, t_0) | x_0 \rangle \\
 &= \int dx_1 \langle x | \hat{U}(t, t_1) | x_1 \rangle \langle x_1 | \hat{U}(t_1, t_0) | x_0 \rangle \\
 &= \int dx_1 iG(x, t; x_1, t_1) \cdot iG(x_1, t_1; x_0, t_0)
 \end{aligned} \tag{1.30}$$

This integral over x_1 represents the “superposition” of all possible “paths” that connect x and x_0 . Next, we try to “smooth” the path in time directly. We can insert more time slices between x and x_0 . If we insert infinitely many time slices, the path becomes smooth.

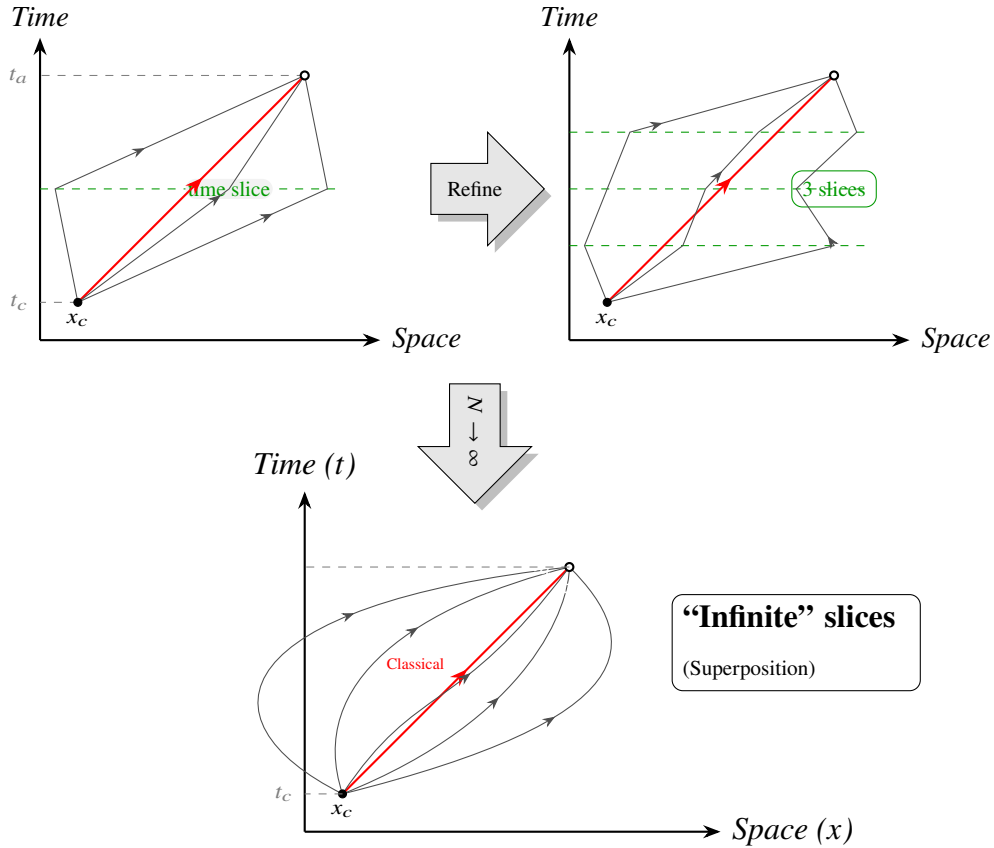


Figure 1: Evolution of the path integral formulation: (Top Left) A single time slice yields coarse, kinked paths. (Top Right) Adding more slices ($N = 3$) refines the grid. (Bottom) Taking the limit of infinite time slices ($N \rightarrow \infty$) recovers the smooth paths of standard quantum mechanics.

First, let us discretize the time domain $[t_0, t]$ into N pieces of equal length $\Delta t = \frac{t-t_0}{N}$:

$$\begin{aligned} iG(x, t; x_0, t_0) &= \langle x | \hat{U}(t, t_{N-1}) \hat{U}(t_{N-1}, t_{N-2}) \cdots \hat{U}(t_1, t_0) | x_0 \rangle \\ &= \int dx_{N-1} \cdots dx_1 \prod_{l=1}^N iG(x_l, t_l; x_{l-1}, t_{l-1}) \end{aligned} \quad (1.31)$$

Let $\mathcal{D}_x = \prod_{l=1}^{N-1} dx_l$. Consider $N \rightarrow \infty$, so $\Delta t = \frac{t-t_0}{N} \rightarrow 0$, which means $t_l - t_{l-1} = \Delta t$.

$$iG(x_l, t_l; x_{l-1}, t_{l-1}) = \langle x_l | e^{-\frac{i}{\hbar} \hat{H} \Delta t} | x_{l-1} \rangle \quad (1.32)$$

Because Δt is small, we can approximate the exponential function by its Taylor series:

$$e^{-\frac{i}{\hbar} \hat{H} \Delta t} \approx \hat{\mathbb{I}} - \frac{i}{\hbar} \hat{H} \Delta t = \hat{\mathbb{I}} - \frac{i}{\hbar} \Delta t \left[\frac{\hat{p}^2}{2m} + V(\hat{x}) \right] \quad (1.33)$$

Substitute (1.33) into (1.32):

$$\begin{aligned} iG(x_l, t_l; x_{l-1}, t_{l-1}) &= \int dp_l \langle x_l | p_l \rangle \langle p_l | \hat{\mathbb{I}} - \frac{i}{\hbar} \Delta t \left[\frac{\hat{p}^2}{2m} + V(\hat{x}) \right] | x_{l-1} \rangle \\ &= \int dp_l \langle x_l | p_l \rangle \langle p_l | x_{l-1} \rangle \left[1 - \frac{i}{\hbar} \left(\frac{p_l^2}{2m} + V(x_{l-1}) \right) \Delta t \right] \end{aligned} \quad (1.34)$$

With the approximation $V(x_l) \approx V(x_{l-1})$:

$$\left[1 - \frac{i}{\hbar} \left(\frac{p_l^2}{2m} + V(x_{l-1}) \right) \Delta t \right] \approx \left(1 - \frac{i}{\hbar} H_l \Delta t \right) \approx e^{-\frac{i}{\hbar} H_l \Delta t} \quad (1.35)$$

Thus, $iG(x_l, t_l; x_{l-1}, t_{l-1})$ can be written as:

$$\begin{aligned} iG(x_l, t_l; x_{l-1}, t_{l-1}) &= \int dp_l \frac{1}{2\pi\hbar} e^{\frac{i}{\hbar} p_l (x_l - x_{l-1})} e^{-\frac{i}{\hbar} H_l \Delta t} \\ &= \frac{1}{2\pi\hbar} \int dp_l e^{\frac{i}{\hbar} [p_l (x_l - x_{l-1}) - H_l \Delta t]} \\ &= \frac{1}{2\pi\hbar} \int dp_l e^{\frac{i}{\hbar} [p_l (\frac{x_l - x_{l-1}}{\Delta t}) - H_l] \Delta t} \end{aligned} \quad (1.36)$$

where H_l is the classical Hamiltonian as a function of p_l and x_l .

When $\Delta t \rightarrow 0$:

$$\frac{x_l - x_{l-1}}{\Delta t} = \dot{x}_l \quad (1.37)$$

Thus, we obtain:

$$p_l \dot{x}_l - H_l = \mathcal{L}_l. \quad (1.38)$$

where $\mathcal{L}_l$ is the classical Lagrangian.

So $iG(x_l, t_l; x_{l-1}, t_{l-1})$ can be written in a Lagrangian form:

$$iG(x_l, t_l; x_{l-1}, t_{l-1}) = \int dp_l \cdot \frac{1}{2\pi\hbar} e^{\frac{i}{\hbar} \mathcal{L}_l \Delta t}. \quad (1.39)$$

Substitute $iG(x_l, t_l; x_{l-1}, t_{l-1})$ into the path integral:

$$\prod_{l=1}^N iG(x_l, t_l; x_{l-1}, t_{l-1}) = \int \frac{dp_N}{2\pi\hbar} \dots \frac{dp_1}{2\pi\hbar} \cdot e^{\frac{i}{\hbar} \sum_{l=1}^N \mathcal{L}_l \Delta t} \quad (1.40)$$

Let $\mathcal{D}_p = \prod_{l=1}^N \frac{dp_l}{2\pi\hbar}$. When $\Delta t \rightarrow 0$, this means:

$$\sum_{l=1}^N \mathcal{L}_l \Delta t = \int_{t_0}^t d\tau \cdot \mathcal{L}[p(\tau), x(\tau)] \quad (1.41)$$

Finally, we can obtain the propagator from the path integral:

Theorem 1.2.1. *Path-integral representation of the propagator:*

$$iG(x, t; x_0, t_0) = \int \mathcal{D}_x \mathcal{D}_p \cdot e^{\frac{i}{\hbar} \int_{t_0}^t d\tau \cdot \mathcal{L}[p(\tau), x(\tau)]} \quad (1.42)$$

where the pair of $p(t)$ and $\dot{x}(t)$ characterizes a path in the px phase space.

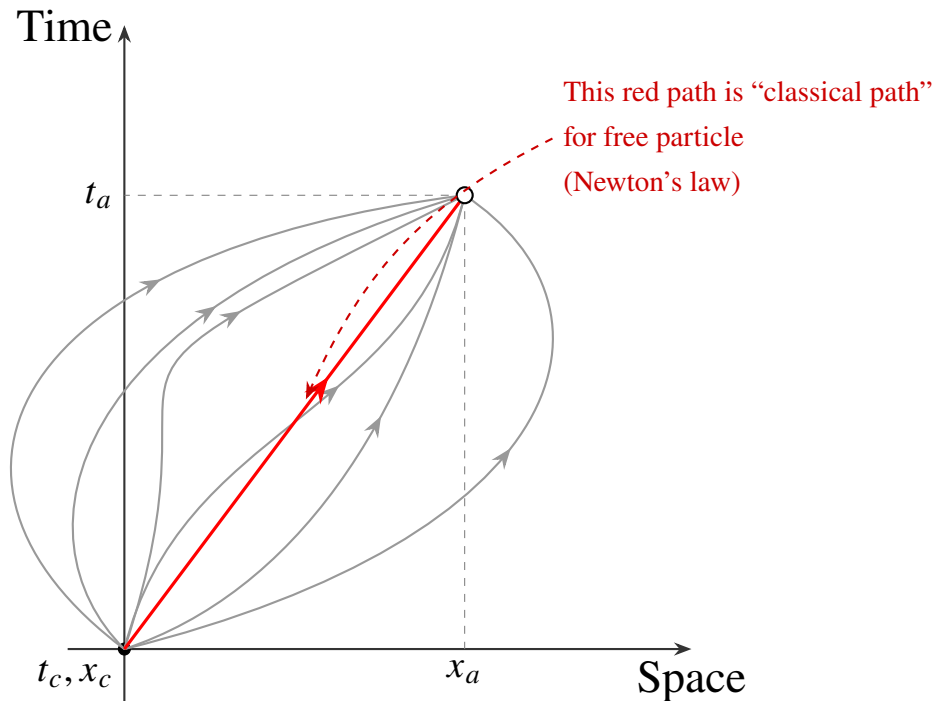


Figure 2: Schematic of Feynman Path Integral. The red path is the classical path (Newton's law) for a free particle, while the gray paths represent the quantum sum over histories.

1.3 Gaussian Integration

If the functional integration over p is Gaussian, we can integrate out p exactly. For example, $H = \frac{p^2}{2m} + V$, so $\mathcal{L} = p\dot{x} - H = p\dot{x} - \frac{p^2}{2m} - V(x)$, and we obtain:

$$iG = \int \mathcal{D}p \mathcal{D}x \exp \left[\frac{i}{\hbar} \sum_t \left(p_t \dot{x}_t - \frac{p_t^2}{2m} - V(x_t) \right) \Delta t \right] \quad (1.43)$$

Let:

$$\mathbf{p} = \begin{pmatrix} p_l \\ \vdots \\ p_1 \end{pmatrix}, \quad \dot{\mathbf{x}} = \begin{pmatrix} \dot{x}_l \\ \vdots \\ \dot{x}_1 \end{pmatrix} \quad (1.44)$$

So we can rewrite the integral as:

$$iG = \int \mathcal{D}x \cdot \exp \left[\frac{i}{\hbar} \sum_{l=1}^N (-V(x_l)) \Delta t \right] \cdot \int \mathcal{D}p \cdot \exp \left[\frac{i}{2m\hbar} (-\mathbf{p}^T \mathbf{p} + 2m\mathbf{p}^T \dot{\mathbf{x}}) \Delta t \right] \quad (1.45)$$

We have a useful formula for the Gaussian integral (Proof in [A](#)):

$$\int \prod_{n=1}^N dx_n \exp \left[-\frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{x}^T \mathbf{y} \right] = (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} \exp \left[\frac{1}{2} \mathbf{y}^T A^{-1} \mathbf{y} \right] \quad (1.46)$$

where $\mathbf{x}, \mathbf{y}$ are real vectors and A is a real symmetric matrix.

Let:

$$\begin{aligned} I &= \int \mathcal{D}p \exp \left[\frac{i}{2m\hbar} (-\mathbf{p}^T \mathbf{p} + 2m\mathbf{p}^T \dot{\mathbf{x}}) \Delta t \right] \\ &= \left(\frac{1}{2\pi\hbar} \right)^N \int \prod_{n=1}^N dp_n \cdot \exp \left[-\frac{1}{2} \mathbf{p}^T A \mathbf{p} - \mathbf{p}^T \dot{\mathbf{x}}' \right] \end{aligned} \quad (1.47)$$

where $A = \frac{i\Delta t}{m\hbar} \mathbb{I}_{N \times N}$ and $\dot{\mathbf{x}}' = -\frac{i\Delta t}{\hbar} \dot{\mathbf{x}}$, $\mathbb{I}_{N \times N}$ is the $N \times N$ identity matrix.

Therefore, we have:

$$\begin{cases} (\det A)^{-\frac{1}{2}} = \left(\frac{i\Delta t}{m\hbar} \right)^{-\frac{N}{2}} \\ A^{-1} = \frac{m\hbar}{i\Delta t} \mathbb{I}_{N \times N} \end{cases} \quad (1.48)$$

The exponent term is:

$$\begin{aligned} \frac{1}{2} \dot{\mathbf{x}}'^T A^{-1} \dot{\mathbf{x}}' &= \frac{1}{2} \cdot \frac{m\hbar}{i\Delta t} \cdot \left(-\frac{i\Delta t}{\hbar} \right)^2 \sum_{l=1}^N \dot{x}_l^2 \\ &= \frac{i}{\hbar} \sum_{l=1}^N \frac{m}{2} \left(\frac{x_l - x_{l-1}}{\Delta t} \right)^2 \Delta t \end{aligned} \quad (1.49)$$

So:

$$\begin{aligned} I &= \left(\frac{1}{2\pi\hbar} \right)^N \cdot (2\pi)^{\frac{N}{2}} \cdot \left(\frac{i\Delta t}{m\hbar} \right)^{-\frac{N}{2}} \cdot \exp \left[\frac{i}{\hbar} \sum_{l=1}^N \frac{m}{2} \left(\frac{x_l - x_{l-1}}{\Delta t} \right)^2 \Delta t \right] \\ &= \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \exp \left[\frac{i}{\hbar} \sum_{l=1}^N \frac{m}{2} \left(\frac{x_l - x_{l-1}}{\Delta t} \right)^2 \Delta t \right] \end{aligned} \quad (1.50)$$

Thus, we can integrate out p to obtain:

$$\begin{aligned} iG &= \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \int \mathcal{D}x \exp \left[\frac{i}{\hbar} \sum_{l=1}^N \frac{m}{2} \left(\frac{x_l - x_{l-1}}{\Delta t} \right)^2 \Delta t - V(x_l)\Delta t \right] \\ &= \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \int \mathcal{D}x \exp \left[\frac{i}{\hbar} \int_{t_0}^t d\tau \mathcal{L}(x, \dot{x}) \right] \end{aligned} \quad (1.51)$$

So the path integral is proportional to:

$$iG \propto \int \mathcal{D}x \cdot e^{\frac{i}{\hbar} S[x(t)]} \quad (1.52)$$

where $S[x(t)] = \int_{t_0}^t dt \mathcal{L}(x, \dot{x})$ is the action and $\mathcal{L}(x, \dot{x}) = \frac{m\dot{x}^2}{2} - V(x)$ is the Lagrangian.

From the path integral in real spacetime, we can gain some intuition about the physical picture:

- (1) Each path is weighted with a $U(1)$ phase factor $e^{\frac{i}{\hbar} S}$. This describes quantum interference between different paths.
- (2) Since $\hbar \sim 10^{-34} \text{ J} \cdot \text{s}$, any "small change" in S (we change S to $S + \delta S$) can lead to strong destructive interference. Therefore, only the paths that satisfy $\delta S = 0$ make dominant contributions to the path integral.
- (3) Remarkably, $\delta S = 0$ is exactly Hamilton's principle in classical mechanics. Thus, the classical paths ($\delta S = 0$) dominate the path integral in the limit $\hbar \rightarrow 0$. In other words, as $\hbar \rightarrow 0$, contributions from other nearby paths cancel out by interference. Therefore, we arrive at the conclusion:

$$\text{Quantum system} \xrightarrow{\hbar \rightarrow 0} \text{classical system}$$

Example 1.3.1. The Hamiltonian of a free particle is $H = \frac{p^2}{2m}$.

Using this Hamiltonian, we can obtain the path integral:

$$iG = \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \int \mathcal{D}x \exp \left[\frac{i}{\hbar} \sum_{l=1}^N \frac{m}{2} \left(\frac{x_l - x_{l-1}}{\Delta t} \right)^2 \Delta t \right]. \quad (1.53)$$

Let:

$$I = \int \mathcal{D}x \exp \left[\frac{im}{2\hbar\Delta t} \sum_{l=1}^N (x_l^2 - 2x_l x_{l-1} + x_{l-1}^2) \right] \quad (1.54)$$

To use the Gaussian integral,

$$\int \prod_{n=1}^N dx_n e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{x}^T \mathbf{y}} = (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} e^{\frac{1}{2} \mathbf{y}^T A^{-1} \mathbf{y}} \quad (1.55)$$

we should rewrite the exponent $\exp[\frac{im}{2\hbar\Delta t} \sum (x_l^2 - 2x_l x_{l-1} + x_{l-1}^2)]$, and define:

$$A = \frac{-im}{\hbar\Delta t} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}_{(N-1) \times (N-1)} \quad (1.56)$$

and:

$$\mathbf{x} = \begin{bmatrix} x_{N-1} \\ \vdots \\ x_1 \end{bmatrix}, \quad \mathbf{y} = \frac{-im}{\hbar\Delta t} \begin{bmatrix} -x_N \\ 0 \\ \vdots \\ \text{all zero} \\ \vdots \\ 0 \\ x_0 \end{bmatrix} \quad (1.57)$$

We get the new form of the exponent term:

$$\exp \left[\frac{im}{2\hbar\Delta t} \sum_{l=1}^{N-1} (x_l^2 - 2x_l x_{l-1} + x_{l-1}^2) \right] = \exp \left(-\frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{x}^T \mathbf{y} \right) e^{\frac{im}{2\hbar\Delta t} (x_N^2 + x_0^2)} \quad (1.58)$$

Because of $\mathcal{D}x = \prod_{l=1}^{N-1} dx_l$ without x_N and x_0 , so:

$$iG = \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \cdot e^{\frac{im}{2\hbar\Delta t} (x_N^2 + x_0^2)} \int \mathcal{D}x e^{-\frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{x}^T \mathbf{y}} \quad (1.59)$$

So we can compute $(\det A) = N \cdot \left(\frac{-im}{\hbar\Delta t} \right)^N$ (Proof in **B**) and get:

$$iG = \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \cdot e^{\frac{im}{2\hbar\Delta t} (x_N^2 + x_0^2)} (2\pi)^{\frac{N}{2}} N^{-1/2} \left(\frac{-im}{\hbar\Delta t} \right)^{-\frac{N}{2}} \cdot e^{\frac{1}{2} \mathbf{y}^T A^{-1} \mathbf{y}} \quad (1.60)$$

Although A^{-1} is difficult to compute, we notice that $\mathbf{y} = \begin{bmatrix} -x_N \\ 0 \\ \vdots \\ 0 \\ x_0 \end{bmatrix}$ has only two non-zero elements, which are located in the first row and the last row, respectively. So we only need

to calculate the first and last columns of matrix A^{-1} , denoted A_1^{-1} and A_{N-1}^{-1} , respectively (Proof in **B**):

$$A_1^{-1} = \frac{i\hbar\Delta t}{mN} \begin{bmatrix} N-1 \\ N-2 \\ \vdots \\ 1 \end{bmatrix}, \quad A_{N-1}^{-1} = \frac{i\hbar\Delta t}{mN} \begin{bmatrix} 1 \\ 2 \\ \vdots \\ N-1 \end{bmatrix} \quad (1.61)$$

The last term of the propagator(1.60) is:

$$\begin{aligned} -\frac{1}{2} \mathbf{y}^T A^{-1} \mathbf{y} &= \frac{1}{2} \mathbf{y}^T [x_N A_1^{-1} + x_0 A_{N-1}^{-1}] \\ &= \frac{im}{2\hbar\Delta t} \left[-(x_N^2 + x_0^2) + \frac{(x_N - x_0)^2}{N} \right] \end{aligned} \quad (1.62)$$

So we can get the complete integral:

$$\begin{aligned} iG &= \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} (2\pi)^{\frac{N}{2}} N^{-\frac{1}{2}} \left(\frac{-im}{\hbar\Delta t} \right)^{-\frac{N-1}{2}} e^{\frac{i}{\hbar} \frac{m(x_N - x_0)^2}{2N\Delta t}} \\ &= \left(\frac{m}{i2\pi\hbar\Delta t N} \right)^{\frac{1}{2}} e^{\frac{i}{\hbar} \frac{m(x_N - x_0)^2}{2N\Delta t}} \end{aligned} \quad (1.63)$$

where $t - t_0 = N\Delta t$, $x = x_N$. So the free particle's propagator is:

$$iG = \left[\frac{m}{i2\pi\hbar(t - t_0)} \right]^{\frac{1}{2}} \cdot e^{\frac{i}{\hbar} \cdot \frac{m(x - x_0)^2}{2(t - t_0)}} \quad (1.64)$$

Chapter 2

Stationary Phase Approximation (Semiclassical Approximation)

We have obtained the propagator iG for a particle to travel from a spacetime point (x_0, t_0) to a final spacetime point (x_f, t_f) , which is given by the Feynman path integral:

$$iG = K(x_f, t_f; x_0, t_0) = \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}} \int \mathcal{D}[x(t)] e^{\frac{i}{\hbar} S[x(t)]} \quad (2.1)$$

where:

- $x(t)$ is the position of the particle at time t , representing a possible path.
- $\mathcal{D}[x(t)]$ is the functional measure for integrating over all paths that satisfy the boundary conditions $x(t_0) = x_0$ and $x(t_f) = x_f$.
- $S[x(t)]$ is the action for a path $x(t)$;

This integral is infinite-dimensional and generally very difficult to calculate directly. So we need an effective method to approximate it. The stationary phase approximation provides a method for approximating it.

2.1 One-dimensional stationary-phase approximation

Consider an integral:

$$I = \int e^{if(x)/a} dx \quad (2.2)$$

where a is a small parameter and $f(x)$ is a real-valued, smooth function.

Our objective is to understand the physical picture of the stationary-phase approximation for the propagator path integral via this one-dimensional stationary-phase approximation.

Let us further define a new function $\Theta(x)$ as:

$$\Theta(x) = \frac{1}{a} f(x) \quad (2.3)$$

$\Theta(x)$ is a phase angle, so:

$$I = \int e^{i\Theta(x)} dx \quad (2.4)$$

This integral can be regarded as an interference experiment.

Because each source at x contributes a phase factor $e^{i\Theta(x)}$, so the total integral I is the result of adding up infinitely many tiny vectors.

In order to compute the integral I , what we really need to do is to find the "dominant contribution" to I .

First, let us pick a point x_0 and evaluate the integral near x_0 . The vicinity of x_0 is given by $x \in (x_0 - \frac{\Delta x}{2}, x_0 + \frac{\Delta x}{2})$. Within this small domain, we may linearize $\Theta(x)$ by a Taylor series:

$$\begin{aligned}\Theta(x) &\approx \Theta(x_0) + \left. \frac{d\Theta}{dx} \right|_{x=x_0} (x - x_0) \\ &= \frac{f(x_0)}{a} + \frac{f'(x_0)}{a} (x - x_0)\end{aligned}\tag{2.5}$$

The contributions to I in the small domain near x_0 are given by:

$$\begin{aligned}I_{x_0}^{\Delta x}(x_0) &= \int_{x_0 - \frac{\Delta x}{2}}^{x_0 + \frac{\Delta x}{2}} dx e^{i\Theta(x_0)} \cdot e^{\frac{if'(x_0)}{a}(x-x_0)} \\ &= e^{i\Theta(x_0)} \int_{x_0 - \frac{\Delta x}{2}}^{x_0 + \frac{\Delta x}{2}} dx e^{\frac{if'(x_0)}{a}(x-x_0)}\end{aligned}\tag{2.6}$$

Let $\Theta(x_0) = \Theta_0$ and $f'(x_0) = f'_0$, so:

$$\begin{aligned}I_{(x_0)}^{\Delta x} &\approx e^{i\Theta_0} \frac{a}{if'_0} \cdot e^{\frac{if'_0}{a}(x-x_0)} \Big|_{x_0 - \frac{\Delta x}{2}}^{x_0 + \frac{\Delta x}{2}} \\ &= e^{i\Theta_0} \frac{2a}{if'_0} \sin \frac{f'_0 \Delta x}{2a}.\end{aligned}\tag{2.7}$$

Let $\alpha = \frac{f'_0 \Delta x}{2a}$, so:

$$I^{\Delta x}(x_0) \approx e^{i\Theta_0} \Delta x \cdot \frac{\sin \alpha}{\alpha}\tag{2.8}$$

Because a is a very small but nonzero parameter (in quantum mechanics, it corresponds to $\hbar$ being a very small but nonzero number), if $f'_0 = 0$, α is strictly equal to 0 and it is not infinitely large near the zero point. So if we consider the case $f'(x_0) = 0$, we get:

$$I^{\Delta x}(x_0) = e^{i\Theta(x_0)} \cdot \Delta x\tag{2.9}$$

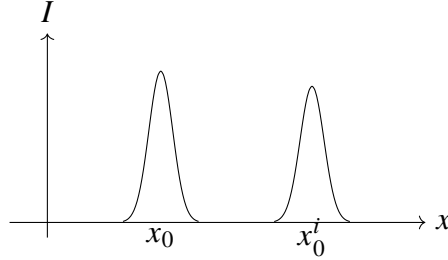
Physically, the result means that all $U(1)$ phases in the vicinity of x_0 are completely the same. It is perfect phase coherence. The superposition of constant phases leads to a linearly enhanced amplitude " Δx ".

Next, let us consider $f'(x_0) \neq 0$. Because a is a very small parameter, $\alpha = \frac{f'_0 \Delta x}{2a}$ is very large. So:

$$\frac{\sin \alpha}{\alpha} \rightarrow 0 \implies I^{\Delta x}(x_0) \approx 0\tag{2.10}$$

Physically, the summation of all U(1) phases in the vicinity of x_0 leads to destructive interference. The smaller a is, the more severe the destructive interference becomes.

In conclusion, if we consider a sufficiently small but nonzero parameter a , it is computationally economical to focus on the contributions from the vicinity of the special points (denoted by a set $\{x_0^i\}$) that satisfy $f'(x_0^i) = 0$.



We consider the quadratic approximation in the vicinity of x_0 :

$$\begin{aligned} f(x) &\approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 \\ &= f(x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 \end{aligned} \quad (2.11)$$

As a result, the original integral I can be evaluated by:

$$\begin{aligned} I &\approx \sum_{\{x_0^i\}} I^{\Delta x}(x_0^i) \\ &= \sum_{\{x_0^i\}} e^{\frac{i}{a}f(x_0^i)} \int_{x_0^i - \frac{\Delta x}{2}}^{x_0^i + \frac{\Delta x}{2}} dx e^{\frac{i}{2a}f''(x_0^i)(x - x_0^i)^2} \end{aligned} \quad (2.12)$$

Because the parameter a is a very small number, far away from the stationary point x_0^i , the contributions of $e^{\frac{i}{2a}f''(x_0^i)(x - x_0^i)^2}$ to the integral cancel each other out through destructive interference.

Therefore, in the above Gaussian integral near x_0 , we can extend the integral bounds to infinity:

$$\begin{aligned} I &\approx \sum_{\{x_0^i\}} e^{\frac{if(x_0^i)}{a}} \int_{-\infty}^{\infty} dx \cdot e^{\frac{i}{2a}f''(x_0^i)(x - x_0^i)^2} \\ &= \sum_{\{x_0^i\}} e^{\frac{if(x_0^i)}{a}} \sqrt{\frac{2\pi ai}{f''(x_0^i)}} \end{aligned} \quad (2.13)$$

In the above integral, we use the Fresnel integral (Proof see Appendix C):

$$\int_{-\infty}^{\infty} e^{ix^2} dx = \sqrt{\frac{\pi}{2}}(1 + i) = \sqrt{\pi}e^{i\pi/4} \quad (2.14)$$

2.2 Semiclassical approximation of Feynman path integrals

We have discussed that if $\hbar$ tends to zero, then the quantum system will transition to the classical system. We only need to treat $f(x)$ as $S[x(t)]$ and the parameter a as $\hbar$ to see the reason, based on the discussion in the previous section. If $\hbar$ is zero, $I^{\Delta x}(x_0)$ equals zero strictly for $x \neq x_0$ and is nonzero only at $x = x_0$. So we only need to consider the classical path with $\delta S = 0$ and do not need to consider the quantum fluctuation near the classical path. But if $\hbar$ is a very small but nonzero number, we need to consider the quantum fluctuation near the classical path. In other words, in classical mechanics, $\hbar = 0$, $I^{\Delta x} \propto \frac{\sin \alpha}{\alpha}$ is equal to zero in $(x_0 - \frac{\Delta x}{2}, x_0) \cup (x_0, x_0 + \frac{\Delta x}{2})$, because $\alpha = \frac{f'_0 \Delta x}{2a}$ tends to infinity no matter how small the radius of this deleted neighborhood is. But in quantum mechanics, $\hbar \sim 10^{-34} \text{J} \cdot \text{s}$, I is not equal to zero in a neighborhood whose radius is on the order of $\hbar$. Therefore, we cannot ignore the contribution from I in this neighborhood.

The Feynman path integral:

$$K(x_f, t_f; x_0, t_0) \propto \int \mathcal{D}[x(t)] \cdot e^{\frac{i}{\hbar} S[x(t)]} \quad (2.15)$$

It can be regarded as the path-integral version of $\int dx e^{\frac{i}{\hbar} f(x)}$. $f(x)$ is replaced by the classical action S_c when $\delta S_c = 0$. Like the previous section, we consider the quadratic approximation in the vicinity of the classical path:

$$\begin{aligned} S &= S_c + \delta S + \frac{1}{2} \delta^2 S \\ &= S_c + \frac{1}{2} \delta^2 S. \end{aligned} \quad (2.16)$$

Now, we can decompose the path near the classical path into the classical path $x_c(t)$ plus a quantum fluctuation $y(t)$ around it:

$$x(t) = x_c(t) + y(t) \quad (2.17)$$

$x(t)$ must satisfy the same boundary conditions, so:

$$\begin{cases} y(t_0) = 0 \\ y(t_f) = 0 \end{cases} \quad (2.18)$$

Let us perform a functional Taylor expansion of the action $S[x(t)]$ around the classical path $x_c(t)$:

$$\begin{aligned} S[x(t)] &= S[x_c + y] = S[x_c] + \int_{t_0}^{t_f} dt \left. \frac{\delta S}{\delta x(t)} \right|_{x=x_c} y(t) \\ &\quad + \frac{1}{2} \int_{t_0}^{t_f} dt_1 \int_{t_0}^{t_f} dt_2 \left. \frac{\delta^2 S}{\delta x(t_1) \delta x(t_2)} \right|_{x=x_c} y(t_1) y(t_2) + O(y^3) \end{aligned} \quad (2.19)$$

In the semiclassical approximation, we assume the fluctuations y are small, so we neglect terms of $O(y^3)$ and higher. At the same time, $\delta S = 0$, so the action $S[x(t)]$:

$$S[x(t)] \approx S_c + \frac{1}{2} \int \int dt_1 dt_2 y(t_1) \cdot \frac{\delta^2 S}{\delta x(t_1) \delta x(t_2)} \Big|_{x=x_c} \cdot y(t_2) \quad (2.20)$$

Now, let's solve the fluctuation term $\delta^2 S$:

$$\delta^2 S = \iint dt_1 dt_2 \cdot y(t_1) \cdot \frac{\delta^2 S}{\delta x(t_1) \delta x(t_2)} \Big|_{x=x_c} \cdot y(t_2) \quad (2.21)$$

Consider a standard Lagrangian $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - V(x)$, so the action is:

$$S = \int dt \left[\frac{1}{2}m\dot{x}^2 - V(x) \right] \quad (2.22)$$

Its second variation is:

$$\delta^2 S = \int dt \cdot \left(\frac{1}{2}m\dot{y}^2 - \frac{1}{2}V''(x_c) \cdot y^2 \right) \quad (2.23)$$

We substitute the approximated action back into the path integral expression:

$$\begin{aligned} K &\propto \int \mathcal{D}[x(t)] \exp \left\{ \frac{i}{\hbar} \left[S_c + \frac{1}{2} \delta^2 S \right] \right\} \\ &= \int \mathcal{D}[x(t)] e^{\frac{i}{\hbar} S_c} \exp \left[\int dt \left(\frac{1}{2}m\dot{y}^2 - \frac{1}{2}V''(x_c)y^2 \right) \right] \end{aligned} \quad (2.24)$$

Now, we need to change the integration variable from an integral over all paths $x(t)$ to an integral over all fluctuations $y(t)$. Since $x_c(t)$ is a fixed classical path, the path measure is:

$$\mathcal{D}[x(t)] = \mathcal{D}[x_c(t) + y(t)] = \mathcal{D}[y(t)] \quad (2.25)$$

and S_c is a constant, so it can be factored out of the integral:

$$\begin{aligned} K &\propto e^{\frac{i}{\hbar} S_c} \cdot \int \mathcal{D}[y(t)] \cdot e^{\frac{i}{2\hbar} \int dt (\frac{1}{2}m\dot{y}^2 - \frac{1}{2}V''(x_c)y^2)} \\ &= F(t_f, t_0) \cdot e^{\frac{i}{\hbar} S_c}. \end{aligned} \quad (2.26)$$

where:

1. $e^{\frac{i}{\hbar} S_c}$ is the Classical Phase Factor. It tells us that in the semiclassical approximation, the evolution of the system's quantum phase is dominated by the classical action. This is a bridge connecting classical and quantum mechanics.
2. $F(t_f, t_0) = \int \mathcal{D}[y(t)] e^{\frac{i}{2\hbar} \delta^2 S}$ is the Quantum Fluctuation Prefactor. It describes the collective contribution of all the small quantum fluctuations around the classical path.

Now let us consider an example: a one-dimensional free particle.

Example 2.2.1. For free particle:

$$\mathcal{L}(x, \dot{x}) = \frac{1}{2}m\dot{x}^2. \quad (2.27)$$

First, we need to calculate the classical action. It is given by the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad (2.28)$$

For free particle:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} \\ \frac{\partial \mathcal{L}}{\partial x} = 0 \end{cases} \quad (2.29)$$

So the equation of motion is:

$$\frac{d}{dt}(m\dot{x}) = m\ddot{x} = 0 \quad (2.30)$$

we can obtain the general solution:

$$x_c(t) = at + b \quad (2.31)$$

We impose the boundary conditions:

$$\begin{cases} x_c(t_0) = x_0 \\ x_c(t_f) = x_f \end{cases} \quad (2.32)$$

to determine a and b :

$$\begin{cases} a = \frac{x_f - x_0}{t_f - t_0} \\ b = x_i - t_i \left(\frac{x_f - x_0}{t_f - t_0} \right) \end{cases} \quad (2.33)$$

The classical action is:

$$S_c = \int_{t_i}^{t_f} dt \cdot \frac{1}{2}m(\dot{x}_c)^2 = \int_{t_i}^{t_f} dt \cdot \frac{1}{2}ma^2 = \frac{m(x_f - x_0)^2}{2(t_f - t_0)} \quad (2.34)$$

So the classical phase factor is:

$$e^{\frac{i}{\hbar}S_c} = e^{\frac{im(x_f - x_0)^2}{2\hbar(t_f - t_i)}} \quad (2.35)$$

Now let us calculate the quantum fluctuation factor:

$$\begin{aligned} F(t_f, t_0) &= \int \mathcal{D}[y(t)] \exp \left\{ \frac{im}{2\hbar} \int_{t_0}^{t_f} \left(\frac{dy}{dt} \right)^2 dt \right\} \\ &= \int \prod_{i=1}^{N-1} dy_i \exp \left[\frac{im}{2\hbar} \sum_{i=1}^N \frac{(y_i - y_{i-1})^2}{\Delta t^2} \Delta t \right] \end{aligned} \quad (2.36)$$

Because $y_N = y_0 = 0$, we can rewrite the sum as:

$$\frac{im}{2\hbar} \sum_{i=1}^N \frac{(y_i - y_{i-1})^2}{\Delta t} = -\frac{1}{2} \mathbf{y}^T M \mathbf{y} \quad (2.37)$$

where

$$M = \frac{m}{i\hbar\Delta t} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}_{(N-1) \times (N-1)}, \quad \mathbf{y} = \begin{pmatrix} y_{N-1} \\ \vdots \\ y_1 \end{pmatrix} \quad (2.38)$$

and use the Gaussian integral:

$$\begin{aligned} F(t_f, t_0) &= (2\pi)^{(N-1)/2} (\det M)^{-1/2} \\ &= (2\pi)^{(N-1)/2} \left(\left(\frac{i\hbar\Delta t}{m} \right)^{N-1} N \right)^{-1/2} \end{aligned} \quad (2.39)$$

So we can get the complete propagator of a free particle:

$$\begin{aligned} K &= J \cdot \left(\frac{2\pi i\hbar\Delta t}{m} \right)^{N/2} \cdot \left(\frac{m}{2\pi i\hbar N\Delta t} \right)^{1/2} \cdot e^{\frac{iS_c}{\hbar}} \\ &= \left(\frac{m}{2\pi i\hbar\Delta t} \right)^{N/2} \cdot \left(\frac{2\pi i\hbar\Delta t}{m} \right)^{N/2} \cdot \left(\frac{m}{2\pi i(t_f - t_0)} \right)^{1/2} \cdot e^{\frac{iS_c}{\hbar}} \\ &= \left(\frac{m}{2\pi i\hbar(t_f - t_0)} \right)^{1/2} \cdot e^{\frac{iS_c}{\hbar}} \end{aligned} \quad (2.40)$$

where, $S_c = \frac{m}{2} \frac{(x_f - x_0)^2}{t_f - t_0}$ and $J = \left(\frac{m}{i2\pi\hbar\Delta t} \right)^{\frac{N}{2}}$ is the coefficient in front of the path integral [2.1](#).

Part II

Quantum Spins, Coherent-state Path Integral, and Topological Terms

Chapter 3

Quantum Spin

We begin by considering the Hilbert space $\mathcal{H}$ for a single quantum spin-1/2 particle. This is a two-dimensional complex vector space.

The conventional approach is to use an orthonormal basis formed by the eigenvectors of the spin operator along a chosen axis, typically the z -axis, denoted $\hat{S}_z$.

$$\hat{S}_z |\uparrow\rangle = +\frac{\hbar}{2} |\uparrow\rangle \quad \text{and} \quad \hat{S}_z |\downarrow\rangle = -\frac{\hbar}{2} |\downarrow\rangle \quad (3.1)$$

Here, $|\uparrow\rangle$ represents the "spin up" state and $|\downarrow\rangle$ represents the "spin down" state. These two states form a complete orthonormal basis, satisfying:

- **Orthogonality:** $\langle\uparrow|\downarrow\rangle = 0$
- **Normalization:** $\langle\uparrow|\uparrow\rangle = \langle\downarrow|\downarrow\rangle = 1$
- **Completeness:** $|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow| = \hat{\mathbb{I}}$

where $\hat{\mathbb{I}}$ is the identity operator in $\mathcal{H}$.

But this simple, discrete basis has two drawbacks:

- This discretely parameterized complete set is not convenient for constructing the path-integral formalism for quantum spins.
- The $SU(2)$ spin symmetry is either explicitly broken or not manifest.

3.1 Spin Coherent States

A general, normalized state $|\psi\rangle$ in the spin-1/2 Hilbert space can be written as a complex linear combination of the basis states:

$$|\psi\rangle = z_1 |\uparrow\rangle + z_2 |\downarrow\rangle \quad (3.2)$$

where $z_1, z_2 \in \mathbb{C}$ are complex coefficients.

The normalization condition $\langle\psi|\psi\rangle = 1$ imposes a constraint on these coefficients:

$$\langle\psi|\psi\rangle = (|z_1|^2 + |z_2|^2) = 1 \quad (3.3)$$

A complex number $z = x + iy$ has two real parameters. Therefore, the pair (z_1, z_2) is defined by four real parameters. The normalization condition $|z_1|^2 + |z_2|^2 = 1$ removes one degree of freedom, leaving three.

Furthermore, in quantum mechanics, the overall phase of a state vector is unphysical. The states $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ (for any real γ) represent the same physical state (i.e., they belong to the same ray in Hilbert space). This "gauge freedom" removes one more degree of freedom.

This leaves $4 - 1 - 1 = 2$ real, physical degrees of freedom. This is a crucial observation: the state space of a spin-1/2 particle is topologically equivalent to the surface of a 2D sphere, which is also parameterized by two angles (like latitude and longitude).

We can explicitly parameterize z_1 and z_2 using two angles, θ and ϕ , which will map directly to the surface of a sphere. A standard (but not unique) parametrization for the spin coherent state, labeled by a unit vector $\mathbf{n}$, is:

$$|\mathbf{n}\rangle \equiv |\theta, \phi\rangle = \cos\left(\frac{\theta}{2}\right) e^{-i\phi/2} |\uparrow\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\phi/2} |\downarrow\rangle \quad (3.4)$$

Here, the spherical coordinate angles have the domains $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. We can easily verify that this state is normalized:

$$\langle\mathbf{n}|\mathbf{n}\rangle = \left|\cos\left(\frac{\theta}{2}\right) e^{-i\phi/2}\right|^2 + \left|\sin\left(\frac{\theta}{2}\right) e^{i\phi/2}\right|^2 = \cos^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right) = 1 \quad (3.5)$$

This set of states $\{|\mathbf{n}\rangle\}$ is continuously parameterized by the angles (θ, ϕ) , addressing the first drawback of the discrete basis.

3.1.1 Physical Interpretation: The Bloch Sphere

To understand the physical meaning of θ and ϕ , we compute the expectation value of the vector spin operator $\hat{\mathbf{S}}$ in the state $|\mathbf{n}\rangle$. We will set $\hbar = 1$ from here on for simplicity. The spin operator is $\hat{\mathbf{S}} = \frac{1}{2}\hat{\boldsymbol{\sigma}}$, where $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$ is the vector of Pauli matrices:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.6)$$

In the $|\uparrow\rangle, |\downarrow\rangle$ basis, $|n\rangle$ is represented by the column vector:

$$|n\rangle = \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix} \implies \langle n| = \begin{pmatrix} \cos(\theta/2) e^{i\phi/2} & \sin(\theta/2) e^{-i\phi/2} \end{pmatrix} \quad (3.7)$$

Expectation value of $\hat{S}_z$:

$$\begin{aligned} \langle \hat{S}_z \rangle &= \langle n| \left(\frac{1}{2} \hat{\sigma}_z \right) |n\rangle = \frac{1}{2} \begin{pmatrix} \cos(\theta/2) e^{i\phi/2} & \sin(\theta/2) e^{-i\phi/2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix} \\ &= \frac{1}{2} \left(\cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right) \right) = \frac{1}{2} \cos(\theta) \end{aligned} \quad (3.8)$$

Expectation value of $\hat{S}_x$:

$$\begin{aligned} \langle \hat{S}_x \rangle &= \langle n| \left(\frac{1}{2} \hat{\sigma}_x \right) |n\rangle = \frac{1}{2} \begin{pmatrix} \dots \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \dots \end{pmatrix} \\ &= \frac{1}{2} \left(\cos(\theta/2) \sin(\theta/2) e^{i\phi/2} e^{i\phi/2} + \sin(\theta/2) \cos(\theta/2) e^{-i\phi/2} e^{-i\phi/2} \right) \\ &= \frac{1}{2} \cos(\theta/2) \sin(\theta/2) \left(e^{i\phi} + e^{-i\phi} \right) = \left(\frac{1}{2} \sin \theta \right) \left(\frac{e^{i\phi} + e^{-i\phi}}{2} \right) = \frac{1}{2} \sin \theta \cos \phi \end{aligned} \quad (3.9)$$

Expectation value of $\hat{S}_y$:

$$\begin{aligned} \langle \hat{S}_y \rangle &= \langle n| \left(\frac{1}{2} \hat{\sigma}_y \right) |n\rangle = \frac{1}{2} \begin{pmatrix} \dots \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \dots \end{pmatrix} \\ &= \frac{1}{2} \left(\cos(\theta/2) (-i) \sin(\theta/2) e^{i\phi/2} e^{i\phi/2} + \sin(\theta/2) (i) \cos(\theta/2) e^{-i\phi/2} e^{-i\phi/2} \right) \\ &= \frac{1}{2} \cos(\theta/2) \sin(\theta/2) \left(-ie^{i\phi} + ie^{-i\phi} \right) = \left(\frac{1}{2} \sin \theta \right) \left(\frac{e^{i\phi} - e^{-i\phi}}{2i} \right) = \frac{1}{2} \sin \theta \sin \phi \end{aligned} \quad (3.10)$$

Combining these results, the expectation value of the spin vector is:

$$\langle \hat{\mathbf{S}} \rangle = \frac{1}{2} (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (3.11)$$

This is a vector of length $S = 1/2$ pointing in the direction specified by the unit vector $\mathbf{n}$:

$$\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (3.12)$$

Thus, the state $|n\rangle$ is the quantum state that "points" in the classical direction $\mathbf{n}$. This direction vector lives on the surface of a unit sphere, known as the **Bloch Sphere**.

This formalism treats all directions $\mathbf{n}$ on an equal footing, making the SU(2) rotational symmetry manifest. This addresses the second drawback of the discrete basis.

The eigenvector equation:

$$(\hat{\mathbf{S}} \cdot \mathbf{n})|n\rangle = \frac{1}{2}|n\rangle \quad (3.13)$$

This equation signifies that the coherent state $|\mathbf{n}\rangle$ is, by definition, the "spin up" eigenvector of the spin operator projected along its own pointing direction $\mathbf{n}$, with the eigenvalue $+1/2$ (with $\hbar = 1$).

Proof. We first construct the operator $\hat{\mathbf{S}} \cdot \mathbf{n}$ in matrix form:

$$\begin{aligned}
 \hat{\mathbf{S}} \cdot \mathbf{n} &= \hat{S}_x n_x + \hat{S}_y n_y + \hat{S}_z n_z \\
 &= \frac{1}{2} (\hat{\sigma}_x n_x + \hat{\sigma}_y n_y + \hat{\sigma}_z n_z) \\
 &= \frac{1}{2} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin \theta \cos \phi + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin \theta \sin \phi + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos \theta \right] \\
 &= \frac{1}{2} \begin{pmatrix} \cos \theta & \sin \theta (\cos \phi - i \sin \phi) \\ \sin \theta (\cos \phi + i \sin \phi) & -\cos \theta \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}
 \end{aligned} \tag{3.14}$$

Now, we apply this operator to the coherent state vector $|\mathbf{n}\rangle$:

$$(\hat{\mathbf{S}} \cdot \mathbf{n})|\mathbf{n}\rangle = \frac{1}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix} \tag{3.15}$$

We compute the top and bottom components of the resulting vector separately.

Top component:

$$\begin{aligned}
 &\frac{1}{2} \left[\cos \theta \cos(\theta/2) e^{-i\phi/2} + \sin \theta e^{-i\phi} \sin(\theta/2) e^{i\phi/2} \right] \\
 &= \frac{1}{2} e^{-i\phi/2} [\cos \theta \cos(\theta/2) + \sin \theta \sin(\theta/2)] \\
 &= \frac{1}{2} e^{-i\phi/2} [\cos(\theta - \theta/2)] \quad (\text{using } \cos(A - B) \text{ identity}) \\
 &= \frac{1}{2} \cos(\theta/2) e^{-i\phi/2}
 \end{aligned} \tag{3.16}$$

This is precisely $\frac{1}{2}$ times the top component of $|\mathbf{n}\rangle$.

Bottom component:

$$\begin{aligned}
 &\frac{1}{2} \left[\sin \theta e^{i\phi} \cos(\theta/2) e^{-i\phi/2} - \cos \theta \sin(\theta/2) e^{i\phi/2} \right] \\
 &= \frac{1}{2} e^{i\phi/2} [\sin \theta \cos(\theta/2) - \cos \theta \sin(\theta/2)] \\
 &= \frac{1}{2} e^{i\phi/2} [\sin(\theta - \theta/2)] \quad (\text{using } \sin(A - B) \text{ identity}) \\
 &= \frac{1}{2} \sin(\theta/2) e^{i\phi/2}
 \end{aligned} \tag{3.17}$$

This is precisely $\frac{1}{2}$ times the bottom component of $|\mathbf{n}\rangle$.

Combining both components, we have shown:

$$(\hat{S} \cdot \mathbf{n})|\mathbf{n}\rangle = \frac{1}{2} \begin{pmatrix} \cos(\theta/2)e^{-i\phi/2} \\ \sin(\theta/2)e^{i\phi/2} \end{pmatrix} = \frac{1}{2}|\mathbf{n}\rangle \quad (3.18)$$

This completes the proof. $\square$

3.1.2 Gauge Choice and Topological Singularities

The parametrization in Eq. (3.4) is not unique, and it hides a subtle topological problem.

- **At the North Pole** ($\theta = 0$): The direction $\mathbf{n}$ is $(0, 0, 1)$. The angle ϕ is ill-defined. Our formula gives $|\theta = 0\rangle = \cos(0)e^{-i\phi/2}|\uparrow\rangle + \sin(0)\cdots = e^{-i\phi/2}|\uparrow\rangle$. The state vector itself depends on the meaningless angle ϕ . This is a **singularity**.
- **At the South Pole** ($\theta = \pi$): The direction $\mathbf{n}$ is $(0, 0, -1)$. Our formula gives $|\theta = \pi\rangle = \cos(\pi/2)\cdots + \sin(\pi/2)e^{i\phi/2}|\downarrow\rangle = e^{i\phi/2}|\downarrow\rangle$. This is also singular.

This is analogous to the problem of creating a flat map of the Earth: you cannot do so without singularities (e.g., at the poles) or cuts.

We can "fix" the singularity at one pole by making a ϕ -dependent gauge choice (i.e., multiplying by an overall phase $e^{i\gamma(\phi)}$).

Choice 1: Regular at North Pole. Let's choose an overall phase $\gamma = \phi/2$. The new state, $|\mathbf{n}\rangle_N$, is:

$$|\mathbf{n}\rangle_N = e^{i\phi/2}|\mathbf{n}\rangle = \cos\left(\frac{\theta}{2}\right)|\uparrow\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}|\downarrow\rangle \quad (3.19)$$

- At the North Pole ($\theta = 0$): $|\mathbf{n}\rangle_N = \cos(0)|\uparrow\rangle + \sin(0)\cdots = |\uparrow\rangle$. This is now regular and well-defined.
- At the South Pole ($\theta = \pi$): $|\mathbf{n}\rangle_N = \cos(\pi/2)|\uparrow\rangle + \sin(\pi/2)e^{i\phi}|\downarrow\rangle = e^{i\phi}|\downarrow\rangle$. The singularity has been "pushed" to the South Pole.

Choice 2: Regular at South Pole. Let's choose $\gamma = -\phi/2$. The new state, $|\mathbf{n}\rangle_S$, is:

$$|\mathbf{n}\rangle_S = e^{-i\phi/2}|\mathbf{n}\rangle = \cos\left(\frac{\theta}{2}\right)e^{-i\phi}|\uparrow\rangle + \sin\left(\frac{\theta}{2}\right)|\downarrow\rangle \quad (3.20)$$

This state is regular at the South Pole ($|\mathbf{n}\rangle_S = |\downarrow\rangle$) but singular at the North Pole.

This unavoidable singularity is topological in nature and is the origin of the **Berry Phase**, or the "topological term," in the coherent-state path integral.

3.1.3 Over-Completeness and Orthogonality

The set of all coherent states $\{|n\rangle\}$ for all n on the sphere is an **over-complete** basis. The Hilbert space is only 2-dimensional, but we have an infinite, continuous set of states. This means the states are not, in general, orthogonal.

$$\langle n'|n\rangle \neq 0 \quad \text{for } n' \neq n \text{ and } n' \neq -n \quad (3.21)$$

A special exception, as noted in the text, is for antipodal states.

Next, let us prove that $\langle -n|n\rangle = 0$. The antipodal point $-n$ corresponds to the angles $(\theta', \phi') = (\pi - \theta, \phi + \pi)$.

We write the state $|-n\rangle$ using Eq. (3.4):

$$\begin{aligned} |-n\rangle &= \cos\left(\frac{\pi - \theta}{2}\right) e^{-i(\phi + \pi)/2} |\uparrow\rangle + \sin\left(\frac{\pi - \theta}{2}\right) e^{i(\phi + \pi)/2} |\downarrow\rangle \\ &= \cos\left(\frac{\pi}{2} - \frac{\theta}{2}\right) e^{-i\phi/2} e^{-i\pi/2} |\uparrow\rangle + \sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right) e^{i\phi/2} e^{i\pi/2} |\downarrow\rangle \end{aligned} \quad (3.22)$$

Using $\cos(\pi/2 - x) = \sin(x)$, $\sin(\pi/2 - x) = \cos(x)$, $e^{-i\pi/2} = -i$, and $e^{i\pi/2} = i$:

$$|-n\rangle = \sin\left(\frac{\theta}{2}\right) e^{-i\phi/2} (-i) |\uparrow\rangle + \cos\left(\frac{\theta}{2}\right) e^{i\phi/2} (i) |\downarrow\rangle \quad (3.23)$$

Now we compute the inner product $\langle -n|n\rangle$:

$$\begin{aligned} \langle -n|n\rangle &= \left(i \sin\left(\frac{\theta}{2}\right) e^{i\phi/2} \langle \uparrow| + (-i) \cos\left(\frac{\theta}{2}\right) e^{-i\phi/2} \langle \downarrow| \right) \left(\cos\left(\frac{\theta}{2}\right) e^{-i\phi/2} |\uparrow\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\phi/2} |\downarrow\rangle \right) \\ &= i \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) e^{i\phi/2} e^{-i\phi/2} + (-i) \cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) e^{-i\phi/2} e^{i\phi/2} \\ &= i \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) - i \cos\left(\frac{\theta}{2}\right) \sin\left(\frac{\theta}{2}\right) \\ &= 0 \end{aligned} \quad (3.24)$$

This confirms that antipodal states are orthogonal, as expected. For example, $|n = \hat{z}\rangle = |\uparrow\rangle$ is orthogonal to $|n = -\hat{z}\rangle = |\downarrow\rangle$.

Distinction between $|-n\rangle$ and $-|n\rangle$

It is a common point of confusion to mistake the antipodal state $|-n\rangle$ for the state $-|n\rangle$. We must justify that, in general, $|-n\rangle \neq -|n\rangle$.

From our derivation in the previous section, the antipodal state is:

$$|-n\rangle = i \sin\left(\frac{\theta}{2}\right) e^{-i\phi/2} |\uparrow\rangle - i \cos\left(\frac{\theta}{2}\right) e^{i\phi/2} |\downarrow\rangle \quad (3.25)$$

In contrast, the state $-|\mathbf{n}\rangle$ is:

$$-|\mathbf{n}\rangle = -\cos\left(\frac{\theta}{2}\right)e^{-i\phi/2}|\uparrow\rangle - \sin\left(\frac{\theta}{2}\right)e^{i\phi/2}|\downarrow\rangle \quad (3.26)$$

By simple inspection, these two state vectors are clearly not identical. They are, in fact, orthogonal to each other, as we just proved $\langle -\mathbf{n}|\mathbf{n}\rangle = 0$. If $|\mathbf{n}\rangle$ were equal to $-|\mathbf{n}\rangle$, then we would have $\langle -\mathbf{n}|\mathbf{n}\rangle = \langle -\mathbf{n}| - (-\mathbf{n})\rangle = -1 \cdot \langle -\mathbf{n}|\mathbf{n}\rangle = -1$, which contradicts our result of 0 (unless the state is null, which is not the case).

The state $|\mathbf{n}\rangle$ represents a spin pointing in the *opposite direction* (e.g., spin down), while $-|\mathbf{n}\rangle$ represents the *same physical state* as $|\mathbf{n}\rangle$ but with a phase shift of π (since $e^{i\pi} = -1$).

3.1.4 Completeness Relation

Despite being over-complete, the spin coherent states provide a resolution of the identity operator $\hat{\mathbb{I}}$.

Let's call the integral $J = \int d\Omega |\mathbf{n}\rangle\langle\mathbf{n}| = \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta |\mathbf{n}\rangle\langle\mathbf{n}|$.

1. Integrate over ϕ : The off-diagonal terms depend on $e^{\pm i\phi}$.

$$\int_0^{2\pi} e^{\pm i\phi} d\phi = 0 \quad (3.27)$$

The diagonal terms are independent of ϕ .

$$\int_0^{2\pi} 1 d\phi = 2\pi \quad (3.28)$$

After integrating over ϕ , the matrix J becomes diagonal:

$$J = \int_0^\pi \sin\theta d\theta \begin{pmatrix} 2\pi \cos^2(\theta/2) & 0 \\ 0 & 2\pi \sin^2(\theta/2) \end{pmatrix} \quad (3.29)$$

2. Integrate over θ : Both integrals evaluate to 2π . Thus, the full integral is:

$$J = \int d\Omega |\mathbf{n}\rangle\langle\mathbf{n}| = \begin{pmatrix} 2\pi & 0 \\ 0 & 2\pi \end{pmatrix} = 2\pi \hat{\mathbb{I}} \quad (3.30)$$

Dividing by 2π , we arrive at the completeness relation:

$$\frac{1}{2\pi} \int d\Omega |\mathbf{n}\rangle\langle\mathbf{n}| = \hat{\mathbb{I}} \quad (3.31)$$

This relation is the foundation for the coherent-state path integral. It allows us to insert the identity operator at infinitesimally small time steps, t_j , as an integral over the Bloch sphere: $\hat{\mathbb{I}} = \int \frac{d\Omega_j}{2\pi} |\mathbf{n}_j\rangle\langle\mathbf{n}_j|$. Summing over all paths becomes an integral over all $\mathbf{n}_j$ at all times t_j .

3.2 Coherent-state path integral for spin

Consider the quantum amplitude ($\hbar = 1$):

$$\langle n_f | e^{-i\hat{H}(t_f-t_i)} | n_i \rangle \xrightarrow{i(t_f-t_i) \rightarrow T} \langle n_f | e^{-\hat{H}T} | n_i \rangle \quad (3.32)$$

First, consider the case where $\hat{H}$ describes an isolated spin. We divide the total time interval T into N small slices of duration $\Delta t = \frac{T}{N}$. In the limit $N \rightarrow \infty$ and $\Delta t \rightarrow 0$, the time evolution operator can be written as a product of infinitesimal operators:

$$e^{-i\hat{H}T} = \left(e^{-i\hat{H}\Delta t} \right)^N \quad (3.33)$$

The amplitude is then:

$$\langle n_f | e^{-i\hat{H}\Delta t} \dots e^{-i\hat{H}\Delta t} | n_i \rangle \quad (3.34)$$

We now insert the resolution of the identity operator. Let us label our initial state as $|n_0\rangle = |n_i\rangle$ and final state as $|n_N\rangle = |n_f\rangle$.

$$\begin{aligned} K &= \langle n_N | e^{-i\hat{H}\Delta t} \cdot \left(\int \frac{d\mathbf{n}_{N-1}}{2\pi} |n_{N-1}\rangle \langle n_{N-1}| \right) \cdot e^{-i\hat{H}\Delta t} \dots e^{-i\hat{H}\Delta t} | n_0 \rangle \\ &= \int \left(\prod_{l=1}^{N-1} \frac{d\mathbf{n}_l}{2\pi} \right) \cdot \left(\prod_{l=1}^N \langle n_l | e^{-i\hat{H}\Delta t} | n_{l-1} \rangle \right) \end{aligned} \quad (3.35)$$

Since Δt is small, we can expand the exponential to first order:

$$e^{-i\hat{H}\Delta t} \approx \hat{\mathbb{I}} - i\hat{H}\Delta t \quad (3.36)$$

The matrix element becomes:

$$\begin{aligned} \langle n_l | e^{-i\hat{H}\Delta t} | n_{l-1} \rangle &= \langle n_l | \hat{\mathbb{I}} - i\hat{H}\Delta t | n_{l-1} \rangle \\ &= \langle n_l | n_{l-1} \rangle - i\Delta t \langle n_l | \hat{H} | n_{l-1} \rangle \end{aligned} \quad (3.37)$$

Let us represent the state at time t_l as a small deviation from the state at t_{l-1} .

$$|n_l\rangle \approx |n_{l-1}\rangle + \delta |n_{l-1}\rangle = |n_{l-1}\rangle + \left. \frac{d|n(t)\rangle}{dt} \right|_{t=t_{l-1}} \cdot \Delta t \quad (3.38)$$

The overlap is then:

$$\begin{aligned} \langle n_l | n_{l-1} \rangle &\approx \langle n_{l-1} | n_{l-1} \rangle + \left(\left. \frac{d\langle n(t) |}{dt} \right|_{t=t_{l-1}} \right) |n_{l-1}\rangle \Delta t \\ &= 1 + \left(\left. \frac{d\langle n_l |}{dt} \right|_{t=t_{l-1}} \right) |n_{l-1}\rangle \cdot \Delta t \end{aligned} \quad (3.39)$$

This can be written in more suggestive exponential form for small Δt :

$$\langle n_l | n_{l-1} \rangle \approx e^{\langle \dot{n}_{l-1} | n_{l-1} \rangle \cdot \Delta t} \quad (3.40)$$

where $\langle \dot{n}_{l-1} | = \left. \frac{d\langle n(t) |}{dt} \right|_{t=t_{l-1}}$.

From the normalization condition:

$$\frac{d}{dt} \langle n_l | n_l \rangle = \left(\frac{d}{dt} \langle n_l | \right) | n_l \rangle + \langle n_l | \left(\frac{d}{dt} | n_l \rangle \right) = 0 \quad (3.41)$$

we obtain:

$$\begin{aligned} \langle \dot{n}_l | n_l \rangle &= -\langle n_l | \dot{n}_l \rangle \\ &= -\langle \dot{n}_l | n_l \rangle^* \end{aligned} \quad (3.42)$$

So $\langle \dot{n}_l | n_l \rangle$ is purely imaginary.

Because we only retain first-order small quantities, so the second term of matrix element becomes:

$$\begin{aligned} -i\Delta t \langle n_l | \hat{H} | n_{l-1} \rangle &\approx -i\Delta t (\langle n_{l-1} | + \langle \dot{n}_{l-1} | \Delta t) \hat{H} | n_{l-1} \rangle \\ &= -i\langle n_{l-1} | \hat{H} | n_{l-1} \rangle \Delta t + O(\Delta t^2) \\ &\approx -i\langle n_{l-1} | \hat{H} | n_{l-1} \rangle \Delta t \end{aligned} \quad (3.43)$$

Putting everything back into the matrix elements:

$$\begin{aligned} \langle n_l | e^{-i\hat{H}\Delta t} | n_{l-1} \rangle &= 1 + (\langle \dot{n}_{l-1} | n_{l-1} \rangle - i\langle n_{l-1} | \hat{H} | n_{l-1} \rangle) \Delta t \\ &\approx \exp [(\langle \dot{n}_{l-1} | n_{l-1} \rangle - i\langle n_{l-1} | \hat{H} | n_{l-1} \rangle) \Delta t] \end{aligned} \quad (3.44)$$

So the amplitude K is:

$$K = \int \left(\prod_{l=1}^{N-1} \frac{d\mathbf{n}_l}{2\pi} \right) \cdot e^{\sum_{l=1}^N (\langle \dot{n}_{l-1} | n_{l-1} \rangle - i\langle n_{l-1} | \hat{H} | n_{l-1} \rangle) \Delta t} \quad (3.45)$$

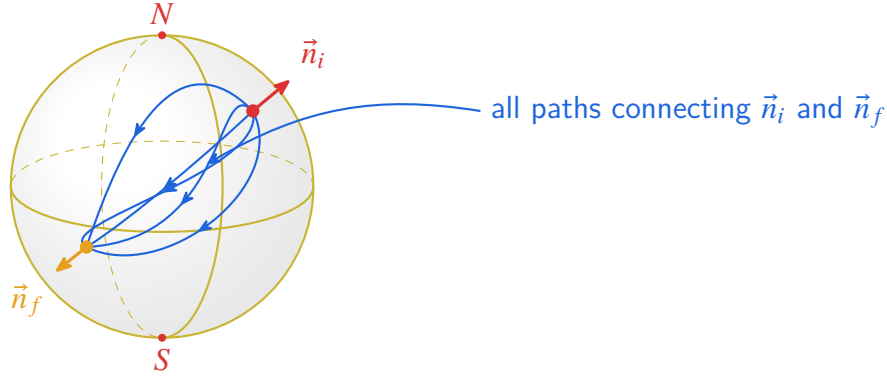
The product of integrals over all intermediate states becomes the formal path integral measure $D[n(t)]$:

$$\begin{aligned} K &= \int D[n(t)] \exp \left\{ i \int_{t_i}^{t_f} [-i\langle \dot{n}(t) | n(t) \rangle - \langle n(t) | \hat{H} | n(t) \rangle] dt \right\} \\ &= \int D[n(t)] \exp \{ iS[n(t)] \} \end{aligned} \quad (3.46)$$

Because of $-\langle \dot{n}(t) | n(t) \rangle = \langle n(t) | \dot{n}(t) \rangle$, $S[n(t)]$ can be written as:

$$S[n(t)] = \int_{t_i}^{t_f} dt \left[\langle n(t) | i \frac{d}{dt} | n(t) \rangle - \langle n(t) | \hat{H} | n(t) \rangle \right] \quad (3.47)$$

where, $\hbar = 1$.



Bloch sphere

Figure 3: The paths connecting $\vec{n}_i$ and $\vec{n}_f$ in the Bloch sphere

3.2.1 Geometrical meaning of the Geometric term

We consider a special system whose Hamiltonian is zero; So the dynamical term vanishes:

$$\langle n(t) | \hat{H} | n(t) \rangle = 0 \quad (3.48)$$

and the action simplifies and contains only the geometric term:

$$S[n(t)] = \int_{t_i}^{t_f} dt \cdot i \langle n(t) | \frac{d}{dt} | n(t) \rangle \quad (3.49)$$

Again, we choose the gauge choice $\delta = \frac{\varphi}{2}$, such that the North Pole is regular:

$$|n(t)\rangle = \cos \frac{\theta(t)}{2} | \uparrow \rangle + e^{i\varphi(t)} \sin \frac{\theta(t)}{2} | \downarrow \rangle \quad (3.50)$$

So the geometric term is:

$$\begin{aligned} \langle n(t) | \frac{d}{dt} | n(t) \rangle &= \cos \frac{\theta}{2} \left(-\frac{1}{2} \sin \frac{\theta}{2} \cdot \dot{\theta} \right) + e^{-i\varphi} \sin \frac{\theta}{2} \cdot (i\dot{\varphi} e^{i\varphi} \sin \frac{\theta}{2} + e^{i\varphi} \frac{1}{2} \cos \frac{\theta}{2} \cdot \dot{\theta}) \\ &= i\dot{\varphi} \sin^2 \frac{\theta}{2} \\ &= i\dot{\varphi} \frac{(1 - \cos \theta)}{2} \end{aligned} \quad (3.51)$$

So the action is:

$$\begin{aligned} S[n(t)] &= -\frac{1}{2} \int_{t_i}^{t_f} dt \cdot \dot{\varphi} (1 - \cos \theta) \\ &= -\frac{1}{2} \int_{\varphi(t_i)}^{\varphi(t_f)} (1 - \cos \theta(\varphi)) d\varphi \end{aligned} \quad (3.52)$$

Because θ and φ depend on t , we can express θ as a function of φ .

We can rewrite the function $1 - \cos \theta(\varphi)$ as an integral:

$$1 - \cos \theta(\varphi) = \int_0^{\theta(\varphi)} d\theta \cdot \sin \theta \quad (3.53)$$

So the action can be written as an area integral:

$$\begin{aligned} S[n(t)] &= -\frac{1}{2} \int_{\varphi_i}^{\varphi_f} d\varphi \int_0^{\theta(\varphi)} \sin \theta d\theta \\ &= -\frac{1}{2} \iint_{\partial A_\gamma} d\Omega \\ &= -\frac{1}{2} A_\gamma \quad (\text{Because the radius is 1 and } A_\gamma = R^2 \Omega) \end{aligned} \quad (3.54)$$

where γ is the path from $|n(t_i)\rangle = |n_i\rangle$ to $|n(t_f)\rangle = |n_f\rangle$ and A_γ is the area enclosed by three points on the sphere ($|\uparrow\rangle, |n_i\rangle, |n_f\rangle$). Obviously, A_γ depends on where the North pole is defined, which is a kind of "gauge dependence". If we multiply $|n(t)\rangle$ by an overall phase:

$$|n(t)\rangle \longrightarrow e^{i\delta} |n(t)\rangle \quad (3.55)$$

$S[n(t)]$ acquires an additional term:

$$\begin{aligned} S'[n(t)] &= i \int_{t_i}^{t_f} dt \langle n(t) | (e^{i\delta} \frac{d}{dt} |n(t)\rangle + i\dot{\delta} e^{i\delta} |n(t)\rangle) \\ &= S[n(t)] + (-1) \int_{t_i}^{t_f} dt \dot{\delta} \\ &= S[n(t)] + (-1)(\delta(t_f) - \delta(t_i)) \end{aligned} \quad (3.56)$$

To be more precise, δ is a function of θ and φ :

$$\delta(t_f) - \delta(t_i) = \delta(\theta_f, \varphi_f) - \delta(\theta_i, \varphi_i) \quad (3.57)$$

If we choose a different gauge, $\Delta\delta$ changes, so $S[n(t)]$ is gauge-dependent.

However, for a closed path ($n_f = n_i$), $S[n(t)]$ is gauge-independent:

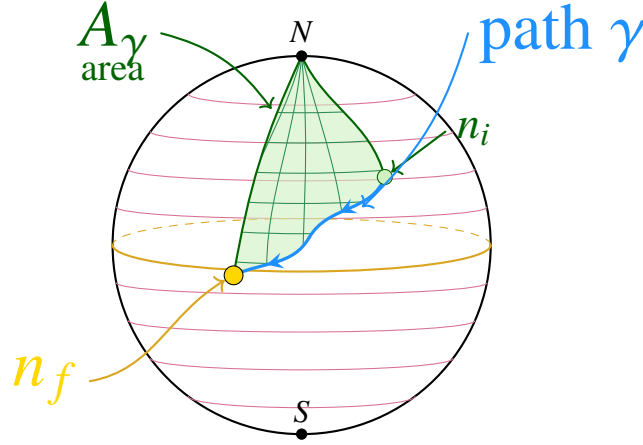
$$\begin{aligned} \Delta\delta &= \delta(\theta_f, \varphi_f) - \delta(\theta_i, \varphi_i) \\ &= 2\pi \cdot N \quad (N \in \mathbb{Z}) \end{aligned} \quad (3.58)$$

The reason is as follows: to ensure that the transformed basis is single-valued, we require

$$e^{i\delta_i} |n_i\rangle \xrightarrow{\text{close path}} e^{i\delta_f} |n_f\rangle = e^{i\delta_i} |n_i\rangle \quad (3.59)$$

and

$$e^{i\delta_f} |n_i\rangle = e^{i\delta_i} |n_i\rangle \quad (3.60)$$

Figure 4: The integral path γ in the Bloch sphere

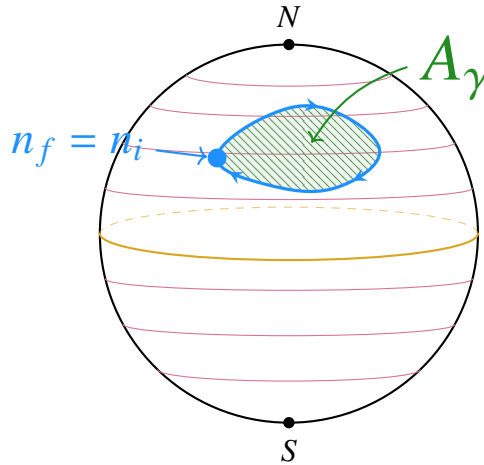
This implies that the value of δ can change by integer multiples of 2π after completing a full cycle:

$$\delta_f - \delta_i = 2\pi \cdot N \quad (N \in \mathbb{Z}). \quad (3.61)$$

So for a closed path, $S[n(t)]$ (more precisely, the geometric phase factor)

$$e^{iS'} = e^{iS+i\Delta\delta} = e^{iS+2\pi N \cdot i} = e^{iS} \quad (3.62)$$

remains unchanged; equivalently, the geometric phase S is invariant modulo 2π .

Figure 5: The integral close path γ in the Bloch sphere

3.2.2 Wess-Zumino term and Witten extension

The Wess-Zumino term is S_{WZ} , which satisfies:

$$S = -\frac{1}{2} \iint d\Omega = \frac{1}{2} S_{\text{WZ}} = S_{\text{spin}} \cdot S_{\text{WZ}}. \quad (3.63)$$

It means that:

$$S_{WZ} = - \iint_{A_\gamma} d\Omega = - \iint_{A_\gamma} \mathbf{n} \cdot d\mathbf{S} \quad (3.64)$$

where $\mathbf{n}$ can be regarded as the monopole field strength with unit charge.

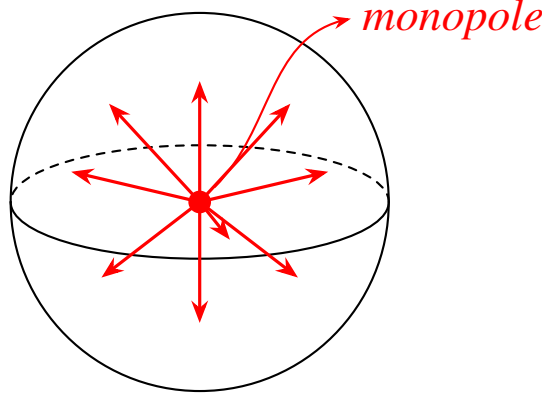


Figure 6: The monopole

Because $\mathbf{n}$ lies on the surface of the sphere, it can be a function of two parameters t and k , which form an orthonormal basis on the sphere (see figure 7). So $d\mathbf{S}$ can be written as:

$$\begin{aligned} d\mathbf{S} &= d\mathbf{n}(t + dt, \kappa) \times d\mathbf{n}(t, \kappa + d\kappa) \\ &= \left(\frac{\partial \mathbf{n}}{\partial t} \cdot dt \right) \times \left(\frac{\partial \mathbf{n}}{\partial \kappa} \cdot d\kappa \right) \\ &= \left(\frac{\partial \mathbf{n}}{\partial t} \times \frac{\partial \mathbf{n}}{\partial \kappa} \right) dt \cdot d\kappa. \end{aligned} \quad (3.65)$$

The Wess-Zumino term has another commonly used expression:

Theorem 3.2.1 (Wess-Zumino term).

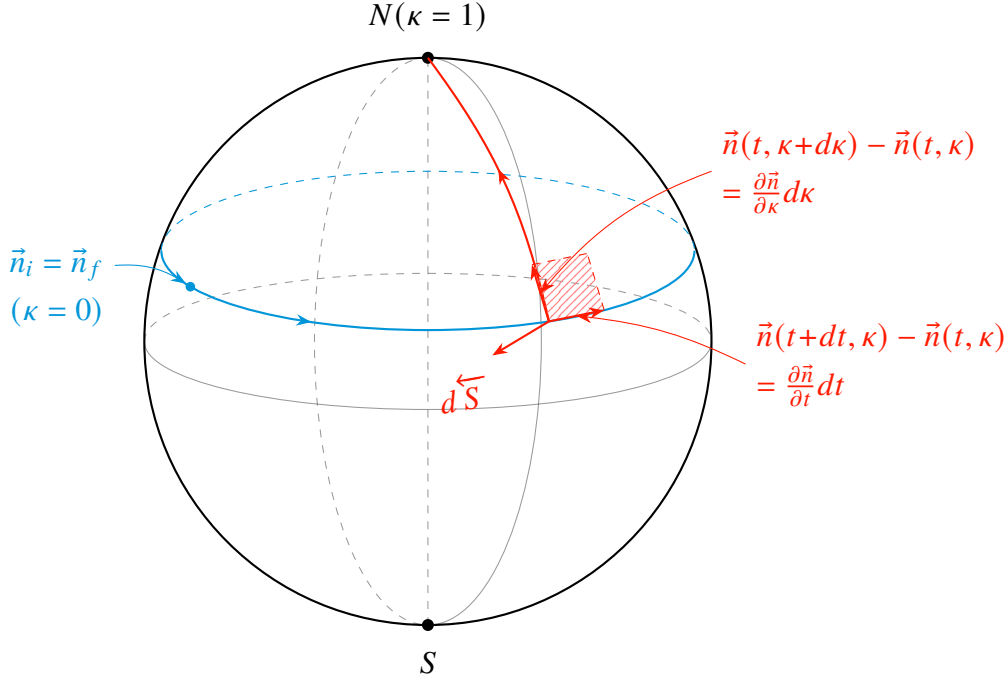
$$S_{WZ} = - \int_0^1 d\kappa \int_{t_i}^{t_f} dt \cdot \mathbf{n}(t, \kappa) \cdot \left[\frac{\partial \mathbf{n}}{\partial t} \times \frac{\partial \mathbf{n}}{\partial \kappa} \right] \quad (3.66)$$

where κ is an auxiliary parameter, in order to extend $\mathbf{n}(t)$ to $\mathbf{n}(t, \kappa)$ with:

$$\begin{cases} \mathbf{n}(t, 0) = \mathbf{n}(t) \\ \mathbf{n}(t, 1) = \mathbf{n}_0(\text{North Pole}) \\ \mathbf{n}(t_i, \kappa) = \mathbf{n}(t_f, \kappa) \text{ (Closed Path)} \end{cases} \quad (3.67)$$

3.2.3 Emergence of U(1) gauge degrees of freedom

$$S = i \int_\gamma dt \langle \mathbf{n}(t) | \partial_t | \mathbf{n}(t) \rangle = i \int_\gamma dt \mathbf{z}^\dagger \cdot \partial_t \mathbf{z} \quad (3.68)$$

Figure 7: The differential of the sphere with κ and t .

where, $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = |\mathbf{n}\rangle$. Using the chain rule where $\mathbf{x} = (x_1, x_2)$ are coordinates on Bloch Sphere:

$$S = i \int_{\gamma} z^{\dagger} \frac{\partial z}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{dt} \cdot dt \quad (3.69)$$

Define the Gauge Potential $\mathcal{A}$ as:

$$\mathcal{A} = iz^{\dagger} \cdot \frac{\partial z}{\partial \mathbf{x}} \quad (3.70)$$

Therefore, S can be written as:

$$S = \int_{\gamma} \mathcal{A} \cdot d\mathbf{l} \quad (3.71)$$

where $d\mathbf{l} = \frac{d\mathbf{x}}{dt} \cdot dt = \mathbf{v} \cdot dt$ is the differential of path γ .

We apply a local gauge transformation where the state z picks up a position-dependent phase $\phi(\mathbf{x})$:

$$z \rightarrow z \cdot e^{i\phi} \quad (3.72)$$

So the Gauge Potential change as:

$$\begin{aligned} \mathcal{A}' &= iz^{\dagger} e^{-i\phi} (e^{i\phi} \frac{\partial z}{\partial \mathbf{x}} + z e^{i\phi} \nabla \phi) \\ &= \mathcal{A} + \nabla \phi \end{aligned} \quad (3.73)$$

So the potential $\mathcal{A}$ is not gauge-invariant.

From what we have discussed above, we can know that: A single spin $\xLeftrightarrow{\text{equivalent to}}$ A spherical Landau problem with a monopole.

Because of : (for closed path)

$$S = \oint_{\gamma} \mathcal{A} \cdot d\mathbf{l} \quad (3.74)$$

where $\mathcal{A}$ can be regard as a magnetic potential that is responsible for a “magnetic monopole” at the center of Bloch Ball. Therefore, S_{WZ} can be regarded as a quantum mechanical problem: a charge particle with sufficiently small mass is confined on the unit sphere, and feels electromagnetic field generated by the “magnetic monopole” at the center of the ball.

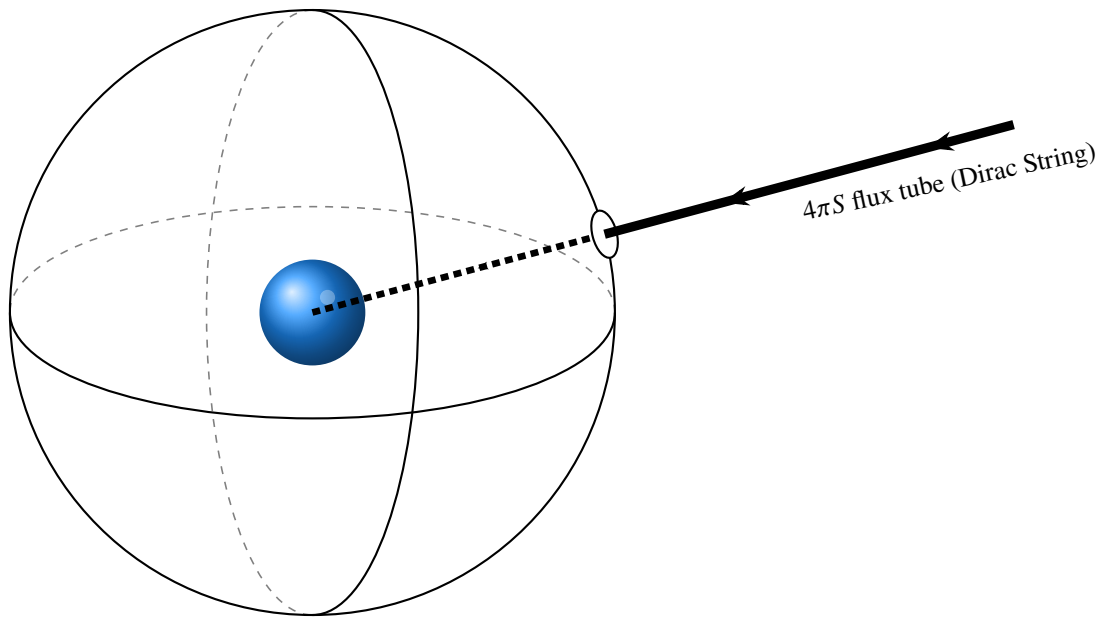


Figure 8: The Dirac String

3.2.4 Conclusion.

In the previous discussion, we have obtained path integral formalism of a single spin- $\frac{1}{2}$ ($S = \frac{1}{2}$).

Theorem 3.2.2 (Action for a generic spin- S). *For a generic spin- S ($S = \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$), the action is given by:*

$$\begin{cases} S[\mathbf{n}(t)] = S_{\text{spin}} \cdot S_{\text{WZ}}[\mathbf{n}(t)] & (\hat{H} = 0) \\ \langle \mathbf{n} | \mathbf{S} | \mathbf{n} \rangle = S_{\text{spin}} \cdot \mathbf{n} \end{cases} \quad (3.75)$$

Chapter 4

The path integral for many-spin system

4.1 General theory

Consider a spin system on a d -dimensional lattice, called the Heisenberg model (figure 9):

$$\hat{H} = J \cdot \sum_{\langle i,j \rangle} \hat{S}_i \cdot \hat{S}_j \quad (4.1)$$

where J is the exchange coupling and $\hat{S}$ are quantum-spin operators. The $\langle i, j \rangle$ denotes a summation over nearest-neighbor pairs.

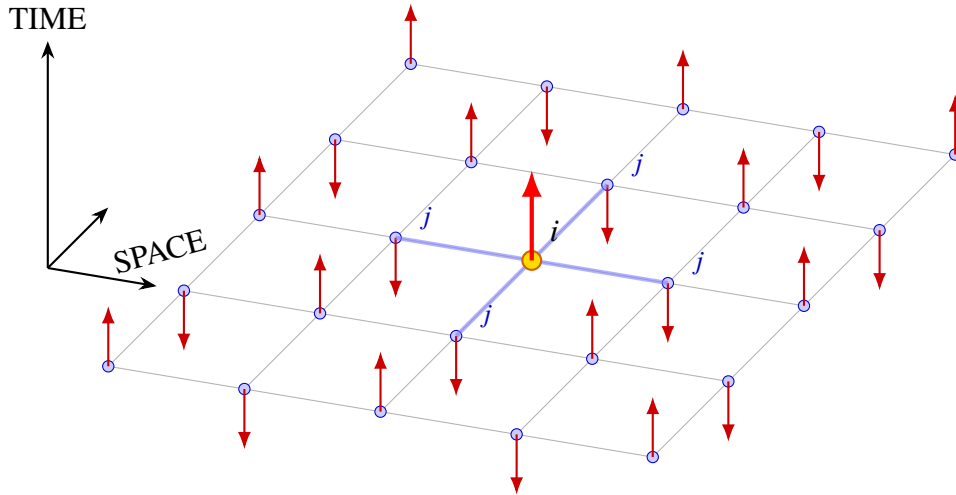


Figure 9: The Heisenberg Model on a 2D square lattice

The partition function of the spin system is:

$$\begin{aligned} Z &= \text{Tre}^{-\beta \hat{H}} \\ &= \int d\mathbf{n}_0 \cdot \langle \mathbf{n}_0 | e^{-\beta \hat{H}} | \mathbf{n}_0 \rangle \end{aligned} \quad (4.2)$$

where $|\mathbf{n}_0\rangle = \bigotimes_{i=1} |\mathbf{n}_0^i\rangle$ is the tensor product of all states on the lattice.

In the previous chapters, we have seen that we can change real time t to imaginary time τ :

$$\langle n_f | e^{-i\hat{H}(t_f-t_i)} | n_i \rangle \xrightarrow{i(t_f-t_i) \rightarrow \tau} \langle n_f | e^{-\hat{H}\tau} | n_i \rangle \quad (4.3)$$

In this version, the propagator K can be written as (for a closed path):

$$K = \int D[\mathbf{n}(t)] e^{-S_E} \quad (4.4)$$

where, $S_E = -iSS_{WZ} = i \cdot S \cdot A_\gamma$ for $\hat{H} = 0$ and $S_E = -iS \cdot S_{WZ} + \int_{\tau_i}^{\tau_f} \langle \mathbf{n}(\tau) | \hat{H} | \mathbf{n}(\tau) \rangle d\tau$ for $\hat{H} \neq 0$.

Because the diagonal element of the partition function and the propagator have the same mathematical form, we can use the same method to evaluate the diagonal elements.

Let $\Delta\tau = \frac{\beta}{N}$ (N is the number of slices), which means that β plays the same role as τ in the diagonal elements, analogous to τ in the propagator ($\tau_i \rightarrow \tau_f \implies 0 \rightarrow \beta$).

But in a many-spin system, the identity operator for the entire system is the tensor product of the identities for each site:

$$\hat{\mathbb{I}}_{total} = \bigotimes_{i=1} \hat{\mathbb{I}}_i = \int \left(\prod_{i=1} d^2 \mathbf{n}_i \right) |\mathbf{n}_1 \cdots \mathbf{n}_N\rangle \langle \mathbf{n}_1 \cdots \mathbf{n}_N| \quad (4.5)$$

So the diagonal element can be written in a form analogous to the propagator:

$$\langle \mathbf{n}_0 | e^{-\beta \hat{H}} | \mathbf{n}_0 \rangle = \int D(\mathbf{n}) \cdot e^{-S_E} \quad (4.6)$$

where $D(\mathbf{n}) = \prod_i \prod_l d^2 \mathbf{n}_i^l$.

Because the diagonal element corresponds to a closed path (i.e., $\langle \mathbf{n}_0 | e^{-\beta \hat{H}} | \mathbf{n}_0 \rangle$, not $\langle \mathbf{n}_i | e^{-\beta \hat{H}} | \mathbf{n}_j \rangle$ for $i \neq j$), S_E can be written as:

$$S_E = \int_0^\beta d\tau \langle \mathbf{n} | \frac{\partial}{\partial \tau} | \mathbf{n} \rangle + \int_0^\beta d\tau \langle \mathbf{n} | \hat{H} | \mathbf{n} \rangle \quad (4.7)$$

For the energy term:

$$\begin{aligned} S_e &= \int_0^\beta d\tau \langle \mathbf{n} | \hat{H} | \mathbf{n} \rangle \\ &= \int_0^\beta d\tau J \sum_{\langle i,j \rangle} (\otimes \langle \mathbf{n} |) \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j (\otimes | \mathbf{n} \rangle) \end{aligned} \quad (4.8)$$

Use the formula:

$$\langle \mathbf{n} | \mathbf{S} | \mathbf{n} \rangle = S_{spin} \mathbf{n} \quad (4.9)$$

we can get:

$$S_e = \int_0^\beta J \sum_{\langle i,j \rangle} S_{spin}^2 \mathbf{n}_i(\tau) \mathbf{n}_j(\tau) \quad (4.10)$$

For the geometric term:

$$\begin{aligned} S_g &= \int_0^\beta d\tau (\otimes \langle \mathbf{n} |) \frac{\partial}{\partial \tau} (\otimes | \mathbf{n} \rangle) \\ &= \int_0^\beta (\otimes \langle \mathbf{n} |) [\otimes (| \mathbf{n} \rangle + d| \mathbf{n} \rangle) - \otimes | \mathbf{n} \rangle] \end{aligned} \quad (4.11)$$

Let's retain the first order term:

$$\begin{aligned}
 S_g &= \int_0^\beta \dots (\otimes \langle \mathbf{n} |) [\otimes | \mathbf{n} \rangle + \sum_i (\bigotimes_{j \neq i} | \mathbf{n}_j \rangle) d| \mathbf{n}_i \rangle] - \otimes | \mathbf{n} \rangle] \\
 &= \sum_i \int_0^\beta (\otimes \langle \mathbf{n} |) (\bigotimes_{j \neq i} | \mathbf{n}_j \rangle) d| \mathbf{n}_i \rangle
 \end{aligned} \tag{4.12}$$

because of $\langle \mathbf{n}_i | \mathbf{n}_i \rangle = 1$, so:

$$\begin{aligned}
 S_g &= \sum_i \int_0^\beta \langle \mathbf{n}_i | d\mathbf{n}_i \rangle \\
 &= \sum_i -iS \cdot S_{WZ}[\mathbf{n}_i] \\
 &= -iS \sum_i S_{WZ}[\mathbf{n}_i]
 \end{aligned} \tag{4.13}$$

let's go back to real time ($\tau \rightarrow it$):

$$\begin{aligned}
 -S_E &\rightarrow \frac{i}{\hbar} S_M \\
 \Rightarrow S_M &= \hbar S \cdot \sum_i S_{WZ}[\mathbf{n}_i] - S^2 \int_0^\beta dt \cdot \sum_{\langle \mathbf{n}_i, \mathbf{n}_j \rangle} J \mathbf{n}_i \cdot \mathbf{n}_j \\
 \frac{iS_M}{\hbar} &= S \{ \sum_i S_{WZ}[\mathbf{n}_i] \} - \frac{S^2}{\hbar} \{ \int_0^\beta dt \sum_{\langle \mathbf{n}_i, \mathbf{n}_j \rangle} J \cdot \mathbf{n}_i \cdot \mathbf{n}_j \}
 \end{aligned} \tag{4.14}$$

So $\{\sum_i S_{WZ}[\mathbf{n}_i]\}$ and $\{\int_0^\beta dt \sum_{\langle i, j \rangle} J \cdot \mathbf{n}_i \cdot \mathbf{n}_j\}$ can be regarded as phase terms in the action ($e^{\frac{i}{\hbar}\theta}$). Therefore, S and $\frac{S^2}{\hbar}$ play the role of $\hbar$ in the simple action. Here, if S is very large, we call it the large- S limit ($\hbar \ll 1$, so $\frac{1}{\hbar} \approx S \gg 1$). This is a semiclassical limit.

4.2 Quantum Ferromagnetism

Let us consider a ferromagnetic system with $J = -|J|$. The action is:

$$S_M = S \sum_i S_{WZ}[\mathbf{n}_i] + |J|S^2 \sum_{\langle i, j \rangle} \int_0^\beta dt \mathbf{n}_i \cdot \mathbf{n}_j \tag{4.15}$$

Since $|\mathbf{n}|^2 = 1$, we can rewrite the term $\mathbf{n}_i \cdot \mathbf{n}_j$ as:

$$\begin{aligned}
 -\mathbf{n}_i \cdot \mathbf{n}_j &= \frac{1}{2}(|\mathbf{n}_i|^2 - 2\mathbf{n}_i \cdot \mathbf{n}_j + |\mathbf{n}_j|^2) - 1 \\
 &= \frac{1}{2}(\mathbf{n}_i - \mathbf{n}_j)^2 - 1
 \end{aligned} \tag{4.16}$$

So the energy term is:

$$S_e = -\frac{|J|S^2}{2} \sum_{\langle i, j \rangle} \int_0^\beta dt (\mathbf{n}_i - \mathbf{n}_j)^2 + |J|S^2 \sum_{\langle i, j \rangle} \int_0^\beta dt \tag{4.17}$$

The second term of S_e is a constant (a global phase), so we can drop it and S_M can be written as:

$$S_M = S \sum_i S_{WZ}[\mathbf{n}_i] - \frac{|J|S^2}{2} \sum_{\langle i,j \rangle} \int_0^\beta dt (\mathbf{n}_i - \mathbf{n}_j)^2 \quad (4.18)$$

Next, we take the continuum limit, where the lattice sums become integrals.

$$\sum_i \rightarrow \int \frac{d^d x}{a_0^d} \quad (4.19)$$

where a_0 is the lattice spacing and finite differences become gradients:

$$\begin{aligned} \sum_j (\mathbf{n}_i - \mathbf{n}_j)^2 &= a_0^2 \sum_j \left(\frac{\Delta \mathbf{n}_j}{\Delta x_j} \right)^2 \\ &= a_0^2 (\nabla \mathbf{n})^2 \end{aligned} \quad (4.20)$$

where $\nabla \mathbf{n} = (\frac{\partial \mathbf{n}}{\partial x}, \dots)$ and not $\nabla \mathbf{n} = (\frac{\partial n_x}{\partial x}, \dots)$

So the continuum action is:

$$\begin{aligned} S_M &= \int \frac{d^d x}{a_0^d} S \cdot S_{WZ}[\mathbf{n}] - \frac{|J|S^2}{2} \int \frac{d^d x}{a_0^d} dt a_0^2 (\nabla \mathbf{n})^2 \\ &= \frac{S}{a_0^d} \int d^d x S_{WZ}[\mathbf{n}] - \frac{|J|S^2}{2a_0^{d-2}} \int d^d x dt (\nabla \mathbf{n})^2 \end{aligned} \quad (4.21)$$

We want to find the classical path that minimizes the action. However, simply varying S_M is incorrect because the variable $\mathbf{n}$ is not free; It is constrained to the surface of a sphere:

$$\mathbf{n}^2 - 1 = 0 \quad (4.22)$$

To handle this, we introduce a Lagrange multiplier field λ . So the total action becomes:

$$S_{\text{tot}} = S_M + \int d^d x dt \cdot \frac{\lambda}{2} (\mathbf{n}^2 - 1) \quad (4.23)$$

Now we can safely solve the variation equation below:

$$\delta S_{\text{tot}} = 0 \quad (4.24)$$

We first need the variation of the Wess-Zumino term δS_{WZ} . Geometrically, the S_{WZ} is proportional to the area swept out by the spin vector on the unit sphere (see figure 10).

From the figure 10, we can see that the variation of this area element $\delta(\Delta A)$ with respect to a small change in path $\delta \mathbf{n}$ is given by:

$$\mathbf{n} \cdot \delta(\Delta A) = \delta \mathbf{n} \times \left(\frac{\partial \mathbf{n}}{\partial t} \right) dt \quad (4.25)$$

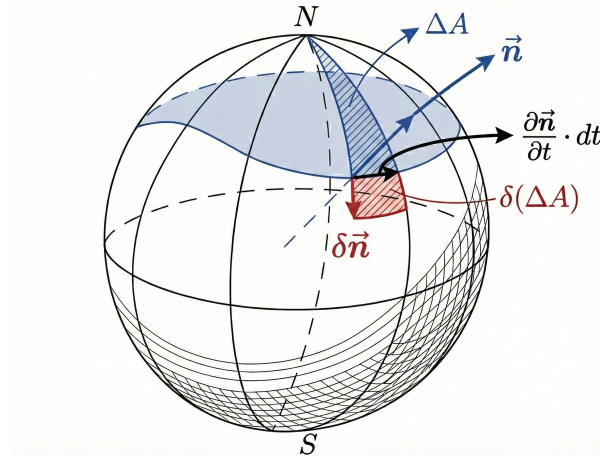


Figure 10: Swz Variation Diagram

We take the cross product with $\mathbf{n}$ on both sides of the equation:

$$\begin{aligned}\delta(\Delta A) &= \mathbf{n}(\delta\mathbf{n} \times \frac{\partial\mathbf{n}}{\partial t})dt \\ &= -\delta\mathbf{n}(\mathbf{n} \times \frac{\partial\mathbf{n}}{\partial t})dt.\end{aligned}\quad (4.26)$$

Therefore, the variation of the Wess-Zumino part of the action is:

$$\delta(\int d^d x S_{WZ}(\mathbf{n})) = \frac{S}{a_0^d} \int dt d^d x [\delta\mathbf{n} \cdot (\mathbf{n} \times \frac{\partial\mathbf{n}}{\partial t})] \quad (4.27)$$

This implies the functional derivative is:

$$\frac{\delta S_g}{\delta\mathbf{n}} = \frac{S}{a_0^d} \cdot (\mathbf{n} \times \frac{\partial\mathbf{n}}{\partial t}) \quad (4.28)$$

Next, we calculate the variation of energy term:

$$\begin{aligned}\delta[\int d^d x dt (\nabla\mathbf{n})^2] &= \int d^d x dt \frac{1}{2} \delta(\frac{\partial\mathbf{n}}{\partial x_i})^2 \\ &= \int d^d x dt \cdot \sum_i \frac{1}{2} \cdot 2(\frac{\partial\mathbf{n}}{\partial x_i}) \cdot \delta(\frac{\partial\mathbf{n}}{\partial x_i})\end{aligned}\quad (4.29)$$

And we use the partial integration method:

$$\begin{aligned}\int d^d x \sum_i 2 \left(\frac{\partial\mathbf{n}}{\partial x_i} \right) \cdot \delta \left(\frac{\partial\mathbf{n}}{\partial x_i} \right) &\stackrel{\frac{\partial\mathbf{n}}{\partial x_i} = \partial_i \mathbf{n}}{=} 2 \int d^d x \sum_i (\partial_i \mathbf{n}) \cdot \partial_i (\delta\mathbf{n}) \\ &= 2 \left\{ \left[\sum_i (\partial_i \mathbf{n}) \cdot \delta\mathbf{n} \right] \Big|_{n_0}^{n_N=n_0} - \int d^d x \sum_i \delta\mathbf{n} \cdot \partial_i^2 \mathbf{n} \right\} \\ &= -2 \int d^d x (\nabla^2 \mathbf{n}) \cdot \delta\mathbf{n}\end{aligned}\quad (4.30)$$

So the variation of energy term is

$$\frac{\delta S_e}{\delta \mathbf{n}} = \frac{|J|S^2}{a_0^{d-2}}(\nabla^2 \mathbf{n}) \quad (4.31)$$

Finally, we calculate the variation of λ term:

$$\delta \int d^d x dt \frac{\lambda}{2}(\mathbf{n}^2 - 1) = \int d^d x dt \cdot \lambda \mathbf{n} \delta \mathbf{n} \quad (4.32)$$

So the variation of λ term is:

$$\frac{\delta S_\lambda}{\delta \mathbf{n}} = \lambda \mathbf{n} \quad (4.33)$$

Now we can get the total action's variation:

$$\delta S_{\text{tot}} = 0 \implies \frac{S}{a_0^d}(\mathbf{n} \times \frac{\partial \mathbf{n}}{\partial t}) + \lambda \mathbf{n} = -\frac{|J|S^2}{a_0^{d-2}}\nabla^2 \mathbf{n}. \quad (4.34)$$

To find the value of the Lagrange multiplier, take the dot product of the above equation with $\mathbf{n}$:

$$\underbrace{\frac{S}{a_0^d} \mathbf{n} \cdot \left(\mathbf{n} \times \frac{\partial \mathbf{n}}{\partial t} \right)}_{\substack{= \frac{S}{a_0^d} \frac{\partial \mathbf{n}}{\partial t} \cdot (\mathbf{n} \times \mathbf{n}) \\ = 0}} + \lambda \underbrace{\mathbf{n} \cdot \mathbf{n}}_{=1} = -\frac{|J|S^2}{a_0^{d-2}}(\mathbf{n} \cdot \nabla^2 \mathbf{n}) \quad (4.35)$$

Thus:

$$\lambda = -\frac{|J|S^2}{a_0^{d-2}}(\mathbf{n} \cdot \nabla^2 \mathbf{n}) \quad (4.36)$$

Now, substitute λ back into the equation:

$$\frac{S}{a_0^d} \cdot (\mathbf{n} \times \frac{\partial \mathbf{n}}{\partial t}) - \left[\frac{|J|S^2}{a_0^{d-2}}(\mathbf{n} \cdot \nabla^2 \mathbf{n}) \right] \mathbf{n} = -\frac{|J|S^2}{a_0^{d-2}}\nabla^2 \mathbf{n} \quad (4.37)$$

Rearrange to group the derivative terms on the right:

$$\frac{S}{a_0^d}(\mathbf{n} \times \frac{\partial \mathbf{n}}{\partial t}) = -\frac{|J|S^2}{a_0^{d-2}}[\nabla^2 \mathbf{n} - (\mathbf{n} \cdot \nabla^2 \mathbf{n}) \cdot \mathbf{n}] \quad (4.38)$$

If we look at: $\mathbf{n} \times (\mathbf{n} \times \nabla^2 \mathbf{n})$:

$$\begin{aligned} \mathbf{n} \times (\mathbf{n} \times \nabla^2 \mathbf{n}) &= \mathbf{n} \cdot (\mathbf{n} \cdot \nabla^2 \mathbf{n}) - \nabla^2 \mathbf{n}(\mathbf{n} \cdot \mathbf{n}) \\ &= \mathbf{n}(\mathbf{n} \cdot \nabla^2 \mathbf{n}) - \nabla^2 \mathbf{n}. \end{aligned} \quad (4.39)$$

So,

$$\frac{S}{a_0^d}(\mathbf{n} \times \partial_t \mathbf{n}) = \frac{|J|S^2}{a_0^{d-2}}[\mathbf{n} \times (\mathbf{n} \times \nabla^2 \mathbf{n})] \quad (4.40)$$

Simplified to:

$$\mathbf{n} \times \frac{\partial \mathbf{n}}{\partial t} = \mathbf{n} \times [|J|S a_0^2(\nabla^2 \mathbf{n})] \quad (4.41)$$

We can get the Landau-Lifshitz equation:

$$\frac{\partial \mathbf{n}}{\partial t} = |J|Sa_0^2(\mathbf{n} \times \nabla^2 \mathbf{n}). \quad (4.42)$$

From this equation, we see that the spins undergo precessional motion with an angular velocity $\mathbf{\Omega}$ given by:

$$\mathbf{\Omega} = -|J|Sa_0^2 \nabla^2 \mathbf{n}. \quad (4.43)$$

The Landau-Lifshitz equations can be solved in the linear regime. Let's parametrize $\mathbf{n}$ by the components:

$$\mathbf{n} = \begin{pmatrix} \sigma \\ \boldsymbol{\pi} \end{pmatrix} \text{ and } \boldsymbol{\pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix} \quad (4.44)$$

where σ and $\pi_i (i = 1, 2)$ satisfy the constraint:

$$\sigma^2 + \boldsymbol{\pi}^2 = 1 \quad (4.45)$$

If the system is ordered, $\sigma \sim 1$ and $|\boldsymbol{\pi}|$ is very small, which means that $\boldsymbol{\pi}$ is a small fluctuation around the direction of σ . In this situation, we can treat σ as a constant:

$$\begin{cases} \sigma \approx 1 \\ \nabla^2 \sigma \approx 0 \end{cases} \quad (4.46)$$

Therefore, we can linearize the Landau-Lifshitz equation:

$$\mathbf{n} \times \nabla^2 \mathbf{n} = \begin{pmatrix} \pi_1 \nabla^2 \pi_2 - \pi_2 \nabla^2 \pi_1 \\ -\sigma \nabla^2 \pi_2 \\ \sigma \nabla^2 \pi_1 \end{pmatrix} \quad (4.47)$$

Keeping only the leading-order term:

$$\mathbf{n} \times \nabla^2 \mathbf{n} = \begin{pmatrix} 0 \\ -\nabla^2 \pi_2 \\ \nabla^2 \pi_1 \end{pmatrix} \quad (4.48)$$

Thus, the linearized Landau-Lifshitz equations are:

$$\begin{cases} \partial_t \pi_1 = -|J|Sa_0^2 \nabla^2 \pi_2 \\ \partial_t \pi_2 = +|J|Sa_0^2 \nabla^2 \pi_1 \end{cases} \quad (4.49)$$

We look for plane-wave solutions (spin waves) of the form:

$$\pi_j(\mathbf{x}, t) = \tilde{\pi}_j e^{i(\mathbf{p} \cdot \mathbf{x} - \omega t)} \quad (4.50)$$

Taking derivatives of this ansatz converts differential operators into algebraic variables:

$$\begin{cases} \partial_t \rightarrow -i\omega \\ \nabla^2 \rightarrow -|\mathbf{p}|^2 \end{cases} \quad (4.51)$$

Substituting these into the linearized equations:

$$\begin{cases} -i\omega\tilde{\pi}_1 = (|J|Sa_0^2|\mathbf{p}|^2)\tilde{\pi}_2 \\ -i\omega\tilde{\pi}_2 = -(|J|Sa_0^2|\mathbf{p}|^2)\tilde{\pi}_1 \end{cases} \quad (4.52)$$

Let $A = |J|Sa_0^2p^2$; the system becomes:

$$\begin{cases} -i\omega\tilde{\pi}_1 - A\tilde{\pi}_2 = 0 \\ A\tilde{\pi}_1 - i\omega\tilde{\pi}_2 = 0 \end{cases} \quad (4.53)$$

For a non-trivial solution ($\tilde{\pi}_1, \tilde{\pi}_2 \neq 0$), the determinant of the coefficient matrix must be zero:

$$\begin{vmatrix} -i\omega & -A \\ A & -i\omega \end{vmatrix} = 0 \quad (4.54)$$

we obtain:

$$\omega = \pm A. \quad (4.55)$$

So we obtain the dispersion relation for ferromagnetic spin waves:

$$|\omega| \approx |J|Sa_0^2|\mathbf{p}|^2 \quad (4.56)$$

We find that the frequency of the low-energy excitations of a quantum ferromagnetic scales as the square of the momentum.

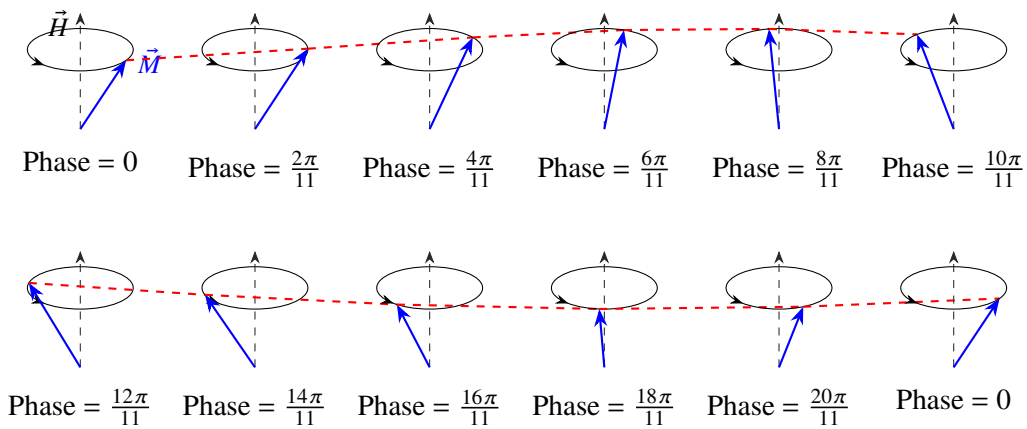


Figure 11: Schematic diagram of spin wave phase variation

4.3 Quantum Antiferromagnetism in One Dimension

Consider a spin chain with an even number of sites N ; we can write down its action ($J = |J|$):

$$S_M[\mathbf{n}] = S \sum_i^N S_{WZ}[\mathbf{n}] - \int dt \sum_i^N JS^2 \mathbf{n}(i, t) \cdot \mathbf{n}(i+1, t) \quad (4.57)$$

where we have assumed periodic boundary conditions.

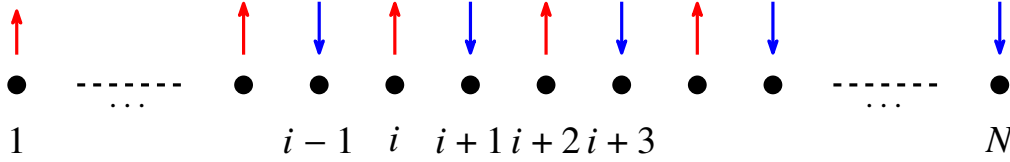


Figure 12: Illustration of a 1D spin chain with an even number of sites, showing alternating spins around site i .

Since we expect to be close to a Néel state, we will stagger the configuration:

$$\mathbf{n}(j) \rightarrow (-1)^j \mathbf{n}(j) \quad (4.58)$$

On a bipartite lattice, the substitution of (4.58) into (4.57) will change the sign of the exchange term of the action to a ferromagnetic:

$$S_M[\mathbf{n}] = S \sum_i^N (-1)^i S_{WZ}[\mathbf{n}(i)] - \frac{|J|S^2}{2} \int dt \sum_{i=1}^N (-1)^i \mathbf{n}(i, t) \cdot (-1)^{i+1} \mathbf{n}(i+1, t) \quad (4.59)$$

so the energy term become to:

$$S_e = \frac{|J|S^2}{2} \int dt \sum_{i=1}^N \mathbf{n}(i, t) \mathbf{n}(i+1, t) \quad (4.60)$$

which is similar to ferromagnetic. But the Wess-Zumino terms become staggered under the replacement of (4.58). Thus, it is the Wess-Zumino term, a purely quantum-mechanical effect, which will distinguish ferromagnets from antiferromagnets.

For an antiferromagnet, we can split $\mathbf{n}$ into a slowly varying piece $\mathbf{m}$ plus a small, rapidly varying part $a_0 \mathbf{l}$, so:

$$\mathbf{n}(i) = \mathbf{m}(i) + (-1)^i a_0 \mathbf{l}(i) \quad (4.61)$$

The constraint $|\mathbf{n}|^2 = 1$ and requirement that $\mathbf{m}$ should obey the same constraint, and demand that $\mathbf{m}$ and $\mathbf{l}$ be orthogonal vector:

$$\begin{cases} |\mathbf{m}|^2 = 1 \\ \mathbf{m} \cdot \mathbf{l} = 0 \end{cases} \quad (4.62)$$

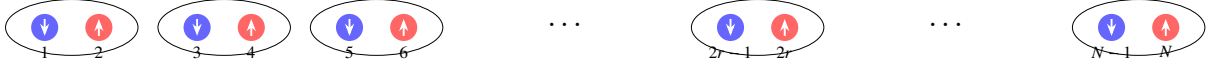


Figure 13: Schematic of 1D Antiferromagnetic Spin Chain

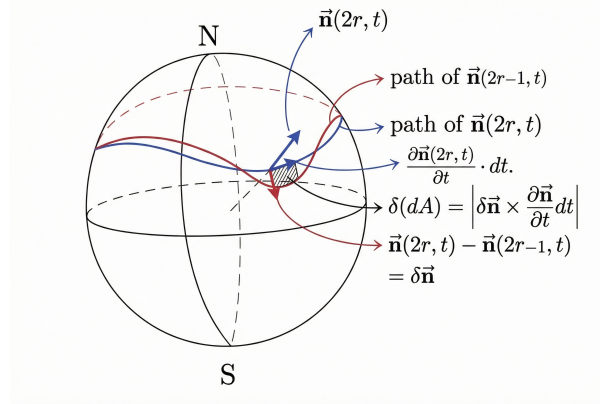


Figure 14: Antiferromagnetic Variation Diagram

The Wess-Zumino terms are rewritten as:

$$S \sum_i^N (-1)^i S_{WZ}[\mathbf{n}(i)] = S \sum_{r=1}^{N/2} \{S_{WZ}[\mathbf{n}(2r)] - S_{WZ}[\mathbf{n}(2r-1)]\} \quad (4.63)$$

By making use of the approximation:

$$\begin{aligned} \mathbf{n}(2r) - \mathbf{n}(2r-1) &= \mathbf{m}(2r) - \mathbf{m}(2r-1) + a_0[\mathbf{l}(2r) + \mathbf{l}(2r-1)] \\ &\approx a_0 \left[\frac{\partial \mathbf{m}(2r)}{\partial x} + 2\mathbf{l}(2r) \right] + O(a_0^2) \end{aligned} \quad (4.64)$$

So the geometric term becomes (use the similar method of area variation, see figure 14)

$$\begin{aligned} S \sum_r^N (-1)^i S_{WZ}[\mathbf{n}(i)] &\approx S \sum_{r=1}^{N/2} (\mathbf{n}(2r) - \mathbf{n}(2r-1)) \left[\mathbf{n}(2r) \times \frac{\partial \mathbf{n}(2r)}{\partial t} \right] \\ &\approx S \sum_{r=1}^{N/2} a_0 \left[\frac{\partial \mathbf{m}(2r)}{\partial x} + 2\mathbf{l}(2r) \right] \left[\mathbf{m}(2r) \times \frac{\partial \mathbf{m}(2r)}{\partial t} \right] \end{aligned} \quad (4.65)$$

Using continuum limit ($a_0 \rightarrow 0$):

$$\begin{aligned} \lim_{a_0 \rightarrow 0} S_g &\approx \frac{S}{2} \sum_{r=1}^N (\dots) \quad (\text{periodic boundary conditions}) \\ &= \frac{S}{2} \int dx dt (\partial_x \mathbf{m} + 2\mathbf{l}) \cdot (\mathbf{m} \times \partial_t \mathbf{m}) \end{aligned} \quad (4.66)$$

Thus, it can be simplified to:

$$S_g \approx \frac{S}{2} \int dx dt \cdot \mathbf{m} (\partial_t \mathbf{m} \times \partial_x \mathbf{m}) + S \int dx dt \mathbf{l} (\mathbf{m} \times \partial_t \mathbf{m}) \quad (4.67)$$

This term comes from alternating sum of WZ terms. Similarly, the continuum limit of the energy terms can also be found to be given by:

$$\begin{aligned}
& \lim_{a_0 \rightarrow 0} \frac{|J|S^2}{2} \int dt \sum_{i=1}^N \mathbf{n}(i, t) \mathbf{n}(i+1, t) \\
& \xrightarrow{\text{drop global phase}} \lim_{a_0 \rightarrow 0} \frac{-|J|S^2}{2} \int dt \sum_{i=1}^N [\mathbf{n}(i+1, t) - \mathbf{n}(i, t)]^2 \\
& \approx \frac{-|J|S^2}{2} \int dt \sum_i [(\mathbf{m}(i+1) - \mathbf{m}(i)) + a_0 \mathbf{l}(i+1) + a_0 \mathbf{l}(i)]^2 \\
& \approx \frac{-|J|S^2}{2} \int dt \sum_i a_0^2 \left(\frac{\Delta \mathbf{m}}{a_0} + 2\mathbf{l} \right)^2 \\
& = \frac{-|J|S^2}{2} \int dt a_0 \int dx [(\partial_x \mathbf{m})^2 + \underbrace{4 \frac{\Delta \mathbf{m}}{a_0} \cdot \mathbf{l} + 4\mathbf{l}^2}_{=0}] \\
& = \frac{-|J|S^2 a_0}{2} \int dt dx [(\partial_x \mathbf{m})^2 + 4\mathbf{l}^2]
\end{aligned} \tag{4.68}$$

So the total Lagrangian density involving both the order parameter $\mathbf{m}$ and $\mathbf{l}$:

$$\mathcal{L}_M(\mathbf{m}, \mathbf{l}) = -2a_0|J|S^2\mathbf{l}^2 + S\mathbf{l}(\mathbf{m} \times \partial_t \mathbf{m}) - \frac{a_0|J|S^2}{2}(\partial_x \mathbf{m})^2 + \frac{S}{2}\mathbf{m} \cdot (\partial_t \mathbf{m} \times \partial_x \mathbf{m}) \tag{4.69}$$

where $\mathbf{m}$ is the order parameter field and $\mathbf{l}$ roughly represents the average spin field.

The Lagrangian has a quadratic term for $\mathbf{l}$:

$$\mathcal{L}_l = -2a_0|J|S^2\mathbf{l}^2 + S(\mathbf{m} \times \partial_t \mathbf{m})\mathbf{l}. \tag{4.70}$$

To integrate out $\mathbf{l}$, we minimize the action with respect to $\mathbf{l}$:

$$\frac{\partial \mathcal{L}_l}{\partial \mathbf{l}} = -4a_0|J|S^2\mathbf{l} + S(\mathbf{m} \times \partial_t \mathbf{m}) = 0 \tag{4.71}$$

so we solve for $\mathbf{l}$:

$$\mathbf{l} = \frac{1}{4a_0|J|S}(\mathbf{m} \times \partial_t \mathbf{m}) \tag{4.72}$$

Now we substitute the expression for $\mathbf{l}$ found above back into $\mathcal{L}_l$:

$$\begin{aligned}
\mathcal{L}_l^{\text{eff}} &= \frac{1}{8a_0|J|}(\mathbf{m} \times \partial_t \mathbf{m})^2 \quad (\mathbf{m} \perp \partial_t \mathbf{m} \text{ because } \mathbf{m} \text{ is on the sphere}) \\
&= \frac{1}{8a_0|J|}(\partial_t \mathbf{m})^2
\end{aligned} \tag{4.73}$$

Finally, we can combine the new term with the original terms that do not depend on $\mathbf{l}$:

$$\begin{aligned}
\mathcal{L}(\mathbf{m}) &= \frac{1}{8a_0J}(\partial_t \mathbf{m})^2 - \frac{a_0JS^2}{2}(\partial_x \mathbf{m})^2 + \frac{S}{2}\mathbf{m}(\partial_t \mathbf{m} \times \partial_x \mathbf{m}) \\
&= \frac{1}{2g} \left[\frac{1}{v_s}(\partial_t \mathbf{m})^2 - v_s(\partial_x \mathbf{m})^2 \right] + \underbrace{\frac{\theta}{8\pi} \epsilon_{\mu\nu} \mathbf{m}(\partial_\mu \mathbf{m} \times \partial_\nu \mathbf{m})}_{\text{Einstein summation}}
\end{aligned} \tag{4.74}$$

where g and v_s are the coupling constant and the spin-wave velocity, respectively:

$$\begin{cases} g = \frac{2}{S} \\ v_s = 2a_0|J|S \end{cases} \quad (4.75)$$

The last term in (4.74) has topological significance. We have chosen the normalization so that the coupling constant θ is given by:

$$\theta = 2\pi S \quad (4.76)$$

The tensor $\epsilon_{\mu\nu}$ is the usual Levi-Civita antisymmetric tensor in two dimensions ($\mu, \nu = t, x$ ($t \rightarrow 0, x \rightarrow 1$)):

$$\epsilon_{\mu\nu} = \begin{cases} 1 & (\mu, \nu) = (0, 1) \\ -1 & (\mu, \nu) = (1, 0) \\ 0 & \text{otherwise.} \end{cases} \quad (4.77)$$

The result is the Lagrangian density of non-linear sigma model.

4.4 The role of topological term.

Go back to Euclidean spacetime ($iS_M \rightarrow -S_E$):

$$iS_M = i \int dt dx \mathcal{L}_M[\mathbf{m}(t, x)] = - \int d\tau dx \mathcal{L}_E[\mathbf{m}(\tau, x)] \quad (4.78)$$

So we can get the Lagrangian density $\mathcal{L}_E$ of Euclidean sector ($x_2 = it = ix_0$)

$$\mathcal{L}_E = \frac{1}{2g} [(\partial_1 \mathbf{m})^2 + (\partial_2 \mathbf{m})^2] + \frac{i\theta}{8\pi} \epsilon_{ij} \mathbf{m} \cdot (\partial_i \mathbf{m} \times \partial_j \mathbf{m}) \quad (4.79)$$

where $i, j = 1, 2$.

We now define the Pontryagin index or topological charge (or winding number) Q of the Euclidean space spin configuration $\{\mathbf{m}(x)\}$ by the following expression:

$$\begin{aligned} Q &= \frac{1}{8\pi} \int d^2x \epsilon_{ij} \mathbf{m} \cdot (\partial_i \mathbf{m} \times \partial_j \mathbf{m}) \\ &= \frac{1}{4\pi} \int d^2x \mathbf{m} \cdot (\partial_1 \mathbf{m} \times \partial_2 \mathbf{m}) \end{aligned} \quad (4.80)$$

We impose the boundary condition that the Euclidean action be finite. This is equivalent to the requirement that asymptotically $\mathbf{m}$ becomes a constant vector $\mathbf{m}_0$ at spatial-time infinity:

$$\lim_{|x| \rightarrow \infty} \mathbf{m}(x) = \mathbf{m}_0 \quad (4.81)$$

Thus, the 2D Euclidean space-time is isomorphic to a sphere S_2 since the fields are identified with $\mathbf{m}_0$ at the point of infinity. However, the order parameter manifold ("target space") is also

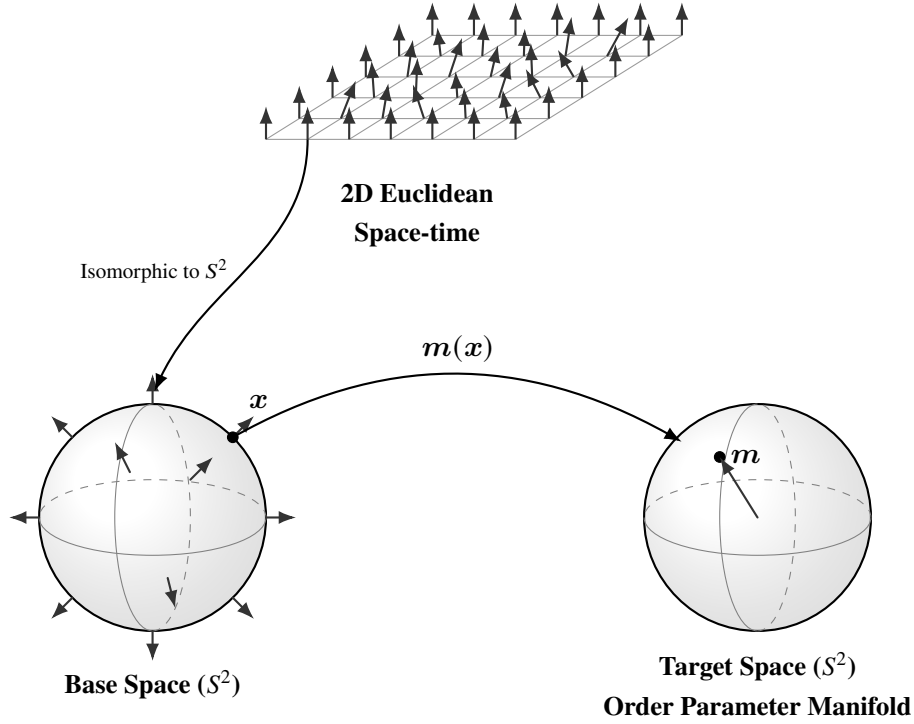


Figure 15: Topological mapping from 2D Euclidean space-time to the base space S^2 and the target space S^2 .

isomorphic to a sphere S_2 , since the constraint $\mathbf{m}^2 = 1$ has to be satisfied everywhere (see figure 15).

The topological charge Q is in the sense that it counts how many times the spin configuration $\mathbf{m}$ has wrapped around the sphere S_2 .

So the topological term of action is:

$$S_{\text{top}} = e^{i \cdot 2\pi S Q} = (-1)^{2S Q} \quad (4.82)$$

Thus:

$$e^{i \cdot 2\pi S Q} = \begin{cases} 1 & \text{for } S = 1, 2, \dots \text{ (integer spin)} \\ (-1)^Q & \text{for } S = \frac{1}{2}, \frac{3}{2}, \dots \text{ (half-odd-integer)} \end{cases} \quad (4.83)$$

We can see that:

- (1) S is an integer: the spin chain is described at low energies by the standard non-linear sigma model, without a topological term.
- (2) S is half integral S : each topological class contributes to the weight of the integral with a sign that is positive (negative) if the winding number Q is even (odd)

The integer- and half-integer-spin chains fall in different universality classes. We will now see that this property implies a very important result, known as Haldane's conjecture, that states that

the integer-spin chains are massive (have an energy gap), whereas the half-integer-spin chains are massless as in the spin one-half case.

Chapter 5

Relativistic Quantum Mechanics.

5.1 Relativistic wave function for Spin-0 particles.

The Klein-Gordon Equation.

The description of phenomena at high energies requires the investigation of relativistic wave equation, which means equations which are invariant under Lorentz transformations. There are several concepts:

1. spatial and temporal coordinates have to be treated equally within the theory.
2. $\Delta x \sim \frac{\hbar}{\Delta p} \sim \frac{\hbar}{m_0 c}$, so a relativistic particle cannot be localized more accurately than $\approx \frac{\hbar}{m_0 c}$. If you try to squeeze a single particle into a space smaller than a certain limit, the uncertainty principle cause its energy to fluctuate so widely that the extra energy to literally create new particle-antiparticle pairs out of the vacuum. Therefore, you can only think of a particle as a single, isolated "free particle" if you don't try to measure its location more precisely than its Compton wavelength $\lambda_c = \frac{\hbar}{m_0 c}$.
3. The time is also uncertain $\Delta t \sim \frac{\Delta x}{c} > \frac{\hbar}{m_0 c^2}$. Because of the finite speed of light, a spatial smear guarantees a temporal smear. You cannot define a perfectly "fixed time t " across the entire spread of the particle.
4. At relativistic energies particle conservation is no longer a valid assumption. A relativistic theory must be able to describe pair creation, vacuum polarization, particle conversion, et.

First we shall remark on the notation used metric tensor:

$$g_{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \quad \text{and} \quad g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \quad (5.1)$$

contravariant four-vector:

$$\begin{cases} x^\mu = \{ct, x, y, z\} \\ x_\mu = \{ct, -x, -y, -z\} \end{cases} \quad (5.2)$$

the indices can be "raised" to give:

$$x^\mu = g^{\mu\nu} x_\nu \quad (5.3)$$

four-momentum vector:

$$p^\mu = \left\{ \frac{E}{c}, p_x, p_y, p_z \right\} \quad (5.4)$$

four-momentum operator:

$$\hat{p}^\mu = i\hbar \frac{\partial}{\partial x_\mu} \equiv i\hbar \nabla^\mu = i\hbar \left\{ \frac{\partial}{\partial(ct)}, -\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}, -\frac{\partial}{\partial z} \right\} \quad (5.5)$$

the commutation relations of momentum and position:

$$\begin{aligned} [\hat{p}^\mu, x^\nu] \psi &= i\hbar \left[\frac{\partial}{\partial x_\mu}, g^{\nu\sigma} x_\sigma \right] \psi \\ &= i\hbar \left[\frac{\partial}{\partial x_\mu} g^{\nu\sigma} x_\sigma - g^{\nu\sigma} x_\sigma \frac{\partial}{\partial x_\mu} \right] \psi \\ &= i\hbar \left(g^{\nu\sigma} x_\sigma \frac{\partial \psi}{\partial x_\mu} + g^{\nu\sigma} \psi \cdot \frac{\partial x_\sigma}{\partial x_\mu} - g^{\nu\sigma} x_\sigma \frac{\partial \psi}{\partial x_\mu} \right) \\ &= i\hbar g^{\nu\sigma} \frac{\partial x_\sigma}{\partial x_\mu} \cdot \psi \\ &= i\hbar g^{\nu\sigma} \delta_\sigma^\mu \psi \\ &= i\hbar g^{\nu\mu} \psi \end{aligned} \quad (5.6)$$

so we can get:

$$[\hat{p}^\mu, x^\nu] = i\hbar g^{\nu\mu} = i\hbar g^{\mu\nu} \quad (5.7)$$

the four-potential of electromagnetic field is given by

$$A^\mu = \{A_0, A_x, A_y, A_z\} \quad (5.8)$$

the electromagnetic field tensor follows in the following way:

$$F^{\mu\nu} = \partial^\nu A^\mu - \partial^\mu A^\nu = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix} \quad (c = 1) \quad (5.9)$$

The Klein-Gordon Equation

The Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi = \left[-\frac{\hbar^2}{2m_0} \nabla^2 + V(x) \right] \psi \quad (5.10)$$

corresponds to the nonrelativistic energy relation in operator forms:

$$\hat{E} = \frac{\hat{p}^2}{2m_0} + V(x) \quad \text{and} \quad \hat{E} = i\hbar \frac{\partial}{\partial t} \quad (5.11)$$

In order to obtain a relativistic wave function we start by considering free particles with the relativistic relation:

$$\begin{aligned} E^2 &= (cp)^2 + (m_0c^2)^2 \\ \Rightarrow p^\mu p_\mu &= (m_0c)^2 \end{aligned} \quad (5.12)$$

We replace the four-momentum p^μ by the four-momentum operator and get the Klein-Gordon equation for free particles:

$$\hat{p}^\mu \hat{p}_\mu \psi = m_0^2 c^2 \psi \quad (5.13)$$

Here m_0 is the rest mass of the particle. We can rewrite it as:

$$\left(\frac{\partial^2}{c^2 \partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} \right) \psi = \frac{m_0^2 c^2}{\hbar^2} \psi \quad (5.14)$$

This is similar to classical wave equation, so the free solution are of the form:

$$\psi = \exp\left(\frac{-i}{\hbar} p_\mu x^\mu\right) = \exp\left[-\frac{i}{\hbar} (Et - \mathbf{p} \cdot \mathbf{x})\right] \quad (5.15)$$

So we have:

$$\begin{aligned} \hat{p}_\mu \hat{p}^\mu \exp\left(\frac{-i}{\hbar} p_\mu x^\mu\right) &= m_0^2 c^2 \psi \\ \Rightarrow p^\mu p_\mu \psi &= m_0^2 c^2 \psi \\ \Rightarrow E^2 &= m_0^2 c^4 + c^2 \mathbf{p}^2 \end{aligned} \quad (5.16)$$

which results in:

$$E = \pm(m_0^2 c^4 + c^2 \mathbf{p}^2)^{\frac{1}{2}}. \quad (5.17)$$

The negative energy are physically connected with antiparticles. Next we construct the four-current j^μ .

$$\begin{cases} (\hat{p}_\mu \hat{p}^\mu - m_0^2 c^2) \psi = 0 & \textcircled{1} \\ (\hat{p}_\mu \hat{p}^\mu - m_0^2 c^2) \psi^* = 0 & \textcircled{2} \end{cases} \quad (5.18)$$

we get $(\psi^* \textcircled{1} - \psi \textcircled{2})$:

$$\begin{aligned} &\psi^* (\hat{p}_\mu \hat{p}^\mu - m_0^2 c^2) \psi - \psi (\hat{p}_\mu \hat{p}^\mu - m_0^2 c^2) \psi^* \\ &= \psi^* \hat{p}_\mu \hat{p}^\mu \psi - \psi \hat{p}_\mu \hat{p}^\mu \psi^* \\ &= -\hbar^2 \psi^* \nabla_\mu \nabla^\mu \psi + \hbar^2 \psi \nabla_\mu \nabla^\mu \psi^* \\ &= \hbar^2 (\psi \nabla_\mu \nabla^\mu \psi^* - \psi^* \nabla_\mu \nabla^\mu \psi) \\ &= 0 \end{aligned} \quad (5.19)$$

we also have :

$$\begin{cases} \nabla_\mu(\psi \nabla^\mu \psi^*) = (\nabla_\mu \psi)(\nabla^\mu \psi^*) + \psi(\nabla_\mu \nabla^\mu \psi^*) \\ \nabla_\mu(\psi^* \nabla^\mu \psi) = (\nabla_\mu \psi^*)(\nabla^\mu \psi) + \psi^*(\nabla_\mu \nabla^\mu \psi) \\ (\nabla_\mu \psi)(\nabla^\mu \psi^*) = (\nabla_\mu \psi^*)(\nabla^\mu \psi) \end{cases} \quad (5.20)$$

So we obtain :

$$\begin{aligned} & \psi \nabla_\mu \nabla^\mu \psi^* - \psi^* \nabla_\mu \nabla^\mu \psi \\ &= \nabla_\mu(\psi \nabla^\mu \psi^* - \psi^* \nabla^\mu \psi) \\ &= \frac{2m_0}{i\hbar} \nabla_\mu j^\mu \\ &= 0 \end{aligned} \quad (5.21)$$

where four-current density is

$$\begin{aligned} j^\mu &= \frac{i\hbar}{2m_0}(\psi \nabla^\mu \psi^* - \psi^* \nabla^\mu \psi) \\ &= \frac{i\hbar}{2m_0} \left\{ \frac{1}{c}(\psi \partial_t \psi^* - \psi^* \partial_t \psi), -(\psi \nabla \psi^* - \psi^* \nabla \psi) \right\} \end{aligned} \quad (5.22)$$

So we have:

$$\begin{aligned} \nabla_\mu j^\mu &= \frac{\partial}{\partial t} \left[\frac{i\hbar}{2m_0 c^2}(\psi \partial_t \psi^* - \psi^* \partial_t \psi) \right] + \nabla \cdot \left[\frac{-i\hbar}{2m_0}(\psi \nabla \psi^* - \psi^* \nabla \psi) \right] \\ &= \frac{\partial}{\partial t} \rho + \nabla \cdot \mathbf{j} \\ &= 0 \end{aligned} \quad (5.23)$$

This is the form of a continuity equation. As usual, integration over the entire configuration space yields:

$$\int_V \frac{\partial \rho}{\partial t} d^3x = \frac{\partial}{\partial t} \int_V \rho d^3x = - \int_V \nabla \cdot \mathbf{j} d^3x = - \int_{\partial V} \mathbf{j} \cdot d\mathbf{s} = 0 \quad (5.24)$$

Hence,

$$\int_V \rho d^3x = \text{const.} \quad (5.25)$$

So it would be a natural guess to interpret ρ as a probability density.

$$\rho = \frac{i\hbar}{2m_0 c^2}(\psi \partial_t \psi^* - \psi^* \partial_t \psi). \quad (5.26)$$

5.2 A Wave Equation for Spin-1/2 Particles - The Dirac Equation

The Schrödinger equation is linear in the time derivative, it is natural to try to construct a Hamiltonian that is also linear in the spatial derivatives (equality of spatial and temporal coordinates). So the desired equation has to be of the form:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi &= \underbrace{\left[-i\hbar c \cdot \left(\hat{\alpha}_1 \frac{\partial}{\partial x_1} + \hat{\alpha}_2 \frac{\partial}{\partial x_2} + \hat{\alpha}_3 \frac{\partial}{\partial x_3} \right) + \hat{\beta} m_0 c^2 \right]}_{=\hat{H}} \psi \quad \textcircled{1} \\ &= \hat{H} \psi \end{aligned} \quad (5.27)$$

We suspect that the $\hat{\alpha}_i$ are matrices and ψ has to be a column vector.

$$\psi = \begin{pmatrix} \psi_1(x, t) \\ \vdots \\ \psi_N(x, t) \end{pmatrix} \quad (5.28)$$

We shall call them spinor and assume its dimension is N . So we have:

$$i\hbar \frac{\partial}{\partial t} \psi_\sigma = \sum_{\tau=1}^N \left[c(\hat{\alpha}_1 \hat{p}_1 + \hat{\alpha}_2 \hat{p}_2 + \hat{\alpha}_3 \hat{p}_3)_{\sigma\tau} + m_0 c^2 \hat{\beta}_{\sigma\tau} \right] \psi_\tau \quad (5.29)$$

To fulfill the correct energy-momentum relation for relativistic free particle:

$$\hat{E}^2 = (\hat{p}c)^2 + (m_0 c^2)^2 \quad (5.30)$$

every single component ψ_σ of the spinor ψ has to satisfy the Klein-Gorden equation:

$$-\hbar^2 \frac{\partial^2}{\partial t^2} \psi_\sigma = (-\hbar^2 c^2 \nabla^2 + m_0^2 c^4) \psi_\sigma \quad (5.31)$$

From equation $\textcircled{1}$, we have :

$$\begin{aligned} -\hbar^2 \frac{\partial^2}{\partial t^2} \psi &= \left(c \sum_i \hat{\alpha}_i \hat{p}_i + m_0 c^2 \hat{\beta} \right) \cdot \left(c \sum_j \hat{\alpha}_j \hat{p}_j + m_0 c^2 \hat{\beta} \right) \psi \\ &= \left[c^2 \sum_i \hat{\alpha}_i^2 \hat{p}_i^2 + c^2 \sum_{i<j} \underbrace{(\hat{\alpha}_i \hat{\alpha}_j \hat{p}_i \hat{p}_j + \hat{\alpha}_j \hat{\alpha}_i \hat{p}_j \hat{p}_i)}_{=(\hat{\alpha}_i \hat{\alpha}_j + \hat{\alpha}_j \hat{\alpha}_i) \hat{p}_i \hat{p}_j} \right. \\ &\quad \left. + m_0 c^3 \sum_i (\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i) \hat{p}_i + \hat{\beta}^2 (m_0 c^2)^2 \right] \psi \end{aligned} \quad (5.32)$$

Comparison with Klein-Gorden equation shows the following requirements for the matrices $\hat{\alpha}_i, \hat{\beta}$:

$$\begin{cases} \hat{\alpha}_i^2 = \hat{\mathbb{I}}, & \hat{\alpha}_i = \hat{\alpha}_i^\dagger \\ \hat{\alpha}_i \hat{\alpha}_j + \hat{\alpha}_j \hat{\alpha}_i = 0, & (i \neq j) \\ \hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i = 0 \\ \hat{\beta}^2 = \hat{\mathbb{I}}, & \hat{\beta} = \hat{\beta}^\dagger \end{cases} \quad (5.33)$$

In order to establish hermiticity of the Hamiltonian $\hat{H}$, the matrices $\hat{\alpha}_i, \hat{\beta}$ also have to be Hermitian:

$$\hat{\alpha}_i = \hat{\alpha}_i^\dagger \quad \text{and} \quad \hat{\beta} = \hat{\beta}^\dagger \quad (5.34)$$

Therefore, the eigenvalues of the matrices are real and can only have the values ± 1 . We assume $\hat{A}_i$ which is the eigenrepresentation of $\hat{\alpha}_i$ has the form:

$$\hat{U}^\dagger \hat{\alpha}_i \hat{U} = \hat{A}_i = \begin{pmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_N \end{pmatrix} \quad (5.35)$$

with the eigenvalues $A_1 \dots A_N$ and $A_j^2 = 1, (j = 1 \dots N)$. So $A_j = \pm 1$

Furthermore, from the anticommutation relation $(\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i = 0)$ it follows that the trace of each $\hat{\alpha}_i$ and $\hat{\beta}$ has to be zero:

$$\begin{aligned} \text{tr}(\hat{\alpha}_i) &= \text{tr}(\hat{\mathbb{I}} \hat{\alpha}_i) \\ &= \text{tr}(\hat{\beta}^2 \hat{\alpha}_i) \quad (\hat{\beta}^2 \hat{\alpha}_i = \hat{\beta} \cdot \hat{\beta} \hat{\alpha}_i, \text{tr} AB = \text{tr} BA) \\ &= \text{tr}(\hat{\beta} \hat{\alpha}_i \hat{\beta}) \quad (\hat{\alpha}_i \hat{\beta} + \hat{\beta} \hat{\alpha}_i = 0 \Rightarrow \hat{\alpha}_i = -\hat{\beta} \hat{\alpha}_i \hat{\beta}^\dagger = -\hat{\beta} \hat{\alpha}_i \hat{\beta}) \\ &= -\text{tr} \hat{\alpha}_i \\ \Rightarrow \text{tr}(\hat{\alpha}_i) &= 0. \end{aligned} \quad (5.36)$$

Similarly, $\text{tr}(\hat{\beta}) = 0$. So we can get:

$$\begin{aligned} \text{tr}(\hat{A}_i) &= \text{tr}(\hat{U}^\dagger \hat{\alpha}_i \hat{U}) = \text{tr}(\hat{\alpha}_i) = 0 \\ \Rightarrow \sum_k A_k &= 0. \end{aligned} \quad (5.37)$$

Because the eigenvalues of $\hat{\alpha}_i$ and $\hat{\beta}$ are equal to ± 1 , each matrix $\hat{\alpha}_i$ and $\hat{\beta}$ has to possess as many positive as negative eigenvalues and therefore has to be of even dimension. The smallest even dimension ($N = 2$) can not be right because only three anticommuting matrices exist (Pauli matrices σ_i). Therefore, the smallest dimension is four. One possible explicit representation of the Dirac matrices are:

$$\hat{\alpha}_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \quad \text{and} \quad \hat{\beta} = \begin{pmatrix} \hat{1} & 0 \\ 0 & -\hat{1} \end{pmatrix} \quad (5.38)$$

where σ_i are Pauli's 2×2 matrices and $\hat{1}$ is the unit matrix

Next we construct the four-current density and the equation of continuity. From Dirac equation and $\psi^\dagger$ we obtain:

$$i\hbar\psi^\dagger \frac{\partial}{\partial t} \psi = \psi^\dagger (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}}) \psi + m_0 c^2 \psi^\dagger \hat{\beta} \psi \quad (5.39)$$

Furthermore, we form the Hermitian conjugate of Dirac equation:

$$-i\hbar \frac{\partial}{\partial t} \psi^\dagger = \psi^\dagger (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}}^\dagger) + m_0 c^2 \psi^\dagger \hat{\beta}^\dagger \quad (5.40)$$

and multiply this equation from the right by ψ to give:

$$-i\hbar \left(\frac{\partial}{\partial t} \psi^\dagger \right) \psi = \psi^\dagger (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}}^\dagger) \psi + m_0 c^2 \psi^\dagger \hat{\beta}^\dagger \psi \quad (5.41)$$

Taking into consideration the hermiticity of the Dirac matrices ($\hat{\alpha}_i^\dagger = \hat{\alpha}_i$, $\hat{\beta}^\dagger = \hat{\beta}$), subtraction of ② from ① yields:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} (\psi^\dagger \psi) &= \psi^\dagger (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}}) \psi - \psi^\dagger (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}}^\dagger) \psi \\ &= -i\hbar (\psi^\dagger (\boldsymbol{\alpha} \cdot c\nabla) \psi) - i\hbar \psi^\dagger (\boldsymbol{\alpha}^\dagger \cdot c\overleftarrow{\nabla}) \psi \\ &= -i\hbar c [\psi^\dagger (\boldsymbol{\alpha} \cdot \nabla) \psi + \psi^\dagger (\boldsymbol{\alpha} \cdot \overleftarrow{\nabla}) \psi] \\ &= -i\hbar c \cdot \nabla \cdot \psi^\dagger \boldsymbol{\alpha} \psi \end{aligned} \quad (5.42)$$

So we can get:

$$\frac{\partial}{\partial t} \rho + \nabla \cdot \mathbf{j} = 0 \quad (5.43)$$

where $\rho = \psi^\dagger \psi$ and $\mathbf{j} = c\psi^\dagger \boldsymbol{\alpha} \psi$

ρ is positive definite and because of the conservation law

$$\frac{\partial}{\partial t} \rho + \nabla \cdot \mathbf{j} = 0 \quad (5.44)$$

we can accept the interpretation of ρ as a probability density. Accordingly, we call $\mathbf{j}$ the probability current density. So $\{c\rho, \mathbf{j}\}$ should form a four-vector:

$$j^\mu = \{c\rho, \mathbf{j}\} \quad (5.45)$$

5.2.1 Free Motion of a Dirac Particle.

The free Dirac equation:

$$i\hbar \frac{\partial}{\partial t} \psi = c \boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + m_0 c^2 \hat{\beta} \psi \quad (5.46)$$

Its stationary states are found with ansatz:

$$\psi(\mathbf{x}, t) = \psi(\mathbf{x}) \exp \left[\frac{-i\epsilon t}{\hbar} \right] \quad (5.47)$$

with which transforms free Dirac equation into:

$$\epsilon \psi(\mathbf{x}) = (c \boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + m_0 c^2 \hat{\beta}) \psi(\mathbf{x}) \quad (5.48)$$

It is useful to split up the four-component spinor into two component spinors ϕ and χ :

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = \begin{pmatrix} \phi \\ \chi \end{pmatrix} \quad \text{with} \quad \begin{cases} \phi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \\ \chi = \begin{pmatrix} \psi_3 \\ \psi_4 \end{pmatrix} \end{cases} \quad (5.49)$$

we can obtain:

$$\epsilon \begin{pmatrix} \phi \\ \chi \end{pmatrix} = c \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix} \cdot \hat{\mathbf{p}} \begin{pmatrix} \phi \\ \chi \end{pmatrix} + m_0 c^2 \begin{pmatrix} \hat{1} & 0 \\ 0 & -\hat{1} \end{pmatrix} \begin{pmatrix} \phi \\ \chi \end{pmatrix} \quad (5.50)$$

$$\Rightarrow \begin{cases} \epsilon \phi = c \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \cdot \chi + m_0 c^2 \phi \\ \epsilon \chi = c \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \phi - m_0 c^2 \chi \end{cases} \quad (5.51)$$

States with definite momentum $\mathbf{p}$ are:

$$\begin{pmatrix} \phi \\ \chi \end{pmatrix} = \begin{pmatrix} \phi_0 \\ \chi_0 \end{pmatrix} \exp \left[\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar} \right] \quad (5.52)$$

So we can get: ($\mathbf{p}$ is the eigenvalues of $\hat{\mathbf{p}}$)

$$\begin{cases} (\epsilon - m_0 c^2) \phi_0 - c \boldsymbol{\sigma} \cdot \mathbf{p} \cdot \chi_0 = 0 \\ -c \boldsymbol{\sigma} \cdot \mathbf{p} \cdot \phi_0 + (\epsilon + m_0 c^2) \cdot \chi_0 = 0 \end{cases} \quad (5.53)$$

This equation has nontrivial solutions only in the case of a vanishing determinant of the coefficients:

$$\begin{vmatrix} (\epsilon - m_0 c^2) \hat{1} & -c \boldsymbol{\sigma} \cdot \mathbf{p} \\ -c \boldsymbol{\sigma} \cdot \mathbf{p} & (\epsilon + m_0 c^2) \hat{1} \end{vmatrix} = 0 \quad (5.54)$$

using the relation:

$$(\boldsymbol{\sigma} \cdot \mathbf{A})(\boldsymbol{\sigma} \cdot \mathbf{B}) = \mathbf{A} \cdot \mathbf{B} \hat{\mathbb{I}} + i\boldsymbol{\sigma} \cdot (\mathbf{A} \times \mathbf{B}) \quad (5.55)$$

we have:

$$\begin{aligned} [\varepsilon^2 - (m_0 c^2)^2] \hat{\mathbb{I}} &= c^2 (\boldsymbol{\sigma} \cdot \mathbf{p})^2 \\ &= c^2 \mathbf{p}^2 \end{aligned} \quad (5.56)$$

Finally, we can get:

$$\varepsilon = \pm E_p \quad \text{with} \quad E_p = c \sqrt{p^2 + m_0^2 c^2} \quad (5.57)$$

From :

$$-c \boldsymbol{\sigma} \cdot \mathbf{p} \phi_0 + (\varepsilon + m_0 c^2) \chi_0 = 0 \quad (5.58)$$

for fixed ε , we can get:

$$\chi_0 = \frac{c(\boldsymbol{\sigma} \cdot \mathbf{p})}{m_0 c^2 + \varepsilon} \phi_0 \quad (5.59)$$

let us denote the two spinor ϕ_0 in the form

$$\phi_0 = U = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad \text{with} \quad U^\dagger U = \hat{\mathbb{I}}. \quad (5.60)$$

We obtain the complete set of positive and negative free solutions of the Dirac equation as:

$$\Psi_{p,\lambda}(\mathbf{x}, t) = N \cdot \left(\frac{U}{\frac{c(\boldsymbol{\sigma} \cdot \mathbf{p})}{m_0 c^2 + \lambda E_p} U} \right) \frac{\exp \left[\frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{x} - \lambda E_p t) \right]}{(2\pi\hbar)^{\frac{3}{2}}}. \quad (5.61)$$

Here $\lambda = \pm 1$ characterizes the positive and negative solutions with time evolution factor $\varepsilon = \lambda E_p$ and N is the normalization factor. So we have:

$$\int \Psi_{p,\lambda}^\dagger \Psi_{p',\lambda'} d^3x = \delta_{\lambda,\lambda'} \delta(\mathbf{p} - \mathbf{p}') \quad (5.62)$$

Hence,

$$N^2 \left[U^\dagger U + U^\dagger \frac{c^2 (\boldsymbol{\sigma} \cdot \mathbf{p})^2}{(m_0 c^2 + \lambda E_p)^2} U \right] = 1 \quad (5.63)$$

using the relation:

$$(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = \mathbf{p}^2 \cdot \hat{\mathbb{I}}, \quad (5.64)$$

we have

$$\begin{aligned}
 N^2 \left[1 + \frac{c^2 p^2}{(m_0 c^2 + \lambda E_p)^2} \right] &= 1 \\
 \Rightarrow N &= \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{c^2 p^2 + (m_0 c^2 + \lambda E_p)^2}} \\
 &= \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{c^2 p^2 + (m_0 c^2)^2 + 2\lambda E_p (m_0 c^2) + (\lambda E_p)^2}} \quad [c^2 p^2 + (m_0 c^2)^2 = (\lambda E_p)^2] \quad (5.65) \\
 &= \sqrt{\frac{(m_0 c^2 + \lambda E_p)^2}{2\lambda E_p (m_0 c^2 + \lambda E_p)}} \\
 &= \sqrt{\frac{(m_0 c^2 + \lambda E_p)}{2\lambda E_p}}
 \end{aligned}$$

For every momentum $\mathbf{p}$, there are two different kinds of solutions, We will now need another quantum number because the energy eigenvalues are two-fold degenerate. It is helicity. First we note that the operator:

$$\boldsymbol{\Sigma} \cdot \mathbf{p} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix} \cdot \mathbf{p} \quad (5.66)$$

and the four-dimensional generalization of the spin vector operator:

$$\hat{S} = \frac{\hbar}{2} \cdot \boldsymbol{\Sigma} \quad (5.67)$$

We calculate

$$\begin{aligned}
 [\hat{H}_f, \boldsymbol{\Sigma} \cdot \mathbf{p}] &= [c\boldsymbol{\alpha} \cdot \mathbf{p} + m_0 c^2 \cdot \hat{\beta}, \boldsymbol{\Sigma} \cdot \mathbf{p}] \\
 &= [c\boldsymbol{\alpha} \cdot \mathbf{p}, \boldsymbol{\Sigma} \cdot \mathbf{p}] + m_0 c^2 \underbrace{[\hat{\beta}, \boldsymbol{\Sigma} \cdot \mathbf{p}]}_{=0} \quad (5.68)
 \end{aligned}$$

Furthermore, we obtain:

$$\begin{aligned}
 [\boldsymbol{\alpha} \cdot \mathbf{p}, \boldsymbol{\Sigma} \cdot \mathbf{p}] &= \boldsymbol{\alpha} \cdot \mathbf{p} \cdot \boldsymbol{\Sigma} \cdot \mathbf{p} - \boldsymbol{\Sigma} \cdot \mathbf{p} \cdot \boldsymbol{\alpha} \cdot \mathbf{p} \quad (\text{simplified notation}) \\
 &= \begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix} - \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix} \begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & (\boldsymbol{\sigma} \cdot \mathbf{p})^2 \\ (\boldsymbol{\sigma} \cdot \mathbf{p})^2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & (\boldsymbol{\sigma} \cdot \mathbf{p})^2 \\ (\boldsymbol{\sigma} \cdot \mathbf{p})^2 & 0 \end{pmatrix} \\
 &= 0
 \end{aligned} \quad (5.69)$$

Hence:

$$[\hat{H}_f, \boldsymbol{\Sigma} \cdot \mathbf{p}] = 0 \quad (5.70)$$

and naturally

$$[\hat{p}, \Sigma \cdot p] = p \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} - \begin{pmatrix} \sigma \cdot p & 0 \\ 0 & \sigma \cdot p \end{pmatrix} p = 0 \quad (5.71)$$

This means that $\Sigma \cdot p$, $\hat{H}_f$ and $\hat{p}$ can be diagonalized together. The same holds for the helicity operator

$$\hat{\Lambda}_S = \frac{\hbar}{2} \frac{\Sigma \cdot p}{|p|} = \hat{S} \cdot \frac{p}{|p|} \quad (5.72)$$

If the electron wave propagates into the direction of z axis, we have:

$$p = \{0, 0, p\} \quad (5.73)$$

so $\hat{\Lambda}_S$ becomes:

$$\hat{\Lambda}_S = \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} \hat{\sigma}_z & 0 \\ 0 & \hat{\sigma}_z \end{pmatrix} \quad (5.74)$$

the eigenvectors of $\hat{\Lambda}_z$ are:

$$\lambda = \frac{\hbar}{2} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}, \quad \lambda = -\frac{\hbar}{2} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\} \quad (5.75)$$

we denote Dirac wave by

$$\left\{ \begin{array}{l} \Psi_{p,\lambda,+\frac{1}{2}} = N \begin{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \frac{c\hat{\sigma}_z p}{m_0 c^2 + \lambda E_p} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} \exp \left[\frac{i}{\hbar} (pz - \lambda E_p t) \right] \\ \Psi_{p,\lambda,-\frac{1}{2}} = N \begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \frac{c\hat{\sigma}_z p}{m_0 c^2 + \lambda E_p} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} \exp \left[\frac{i}{\hbar} (pz - \lambda E_p t) \right] \end{array} \right. \quad (5.76)$$

Part III

Appendix

Appendix A

Multivariate Gaussian Integral

The multivariate Gaussian integral:

$$I = \int \prod_{n=1}^N dx_n \exp \left(-\frac{1}{2} \mathbf{x}^\top A \mathbf{x} - \mathbf{x}^\top \mathbf{y} \right) = (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} \exp \left(\frac{1}{2} \mathbf{y}^\top A^{-1} \mathbf{y} \right) \quad (\text{A.1})$$

where:

- $\mathbf{x}$ and $\mathbf{y}$ are N -dimensional column vectors.
- A is an $N \times N$ real, symmetric, and positive-definite matrix.
- The notation $\int \prod_{n=1}^N dx_n$ denotes integration over all components of $\mathbf{x}$ from $-\infty$ to $+\infty$.

A.1 Proof of the Multivariate Gaussian Integral

The proof relies on the assumptions that A is symmetric ($A^\top = A$) and positive-definite (all eigenvalues are positive), which ensures the integral converges. The proof proceeds in several key steps.

Step 1: Completing the Square

The primary technique is to complete the square for the quadratic form in the exponent. We want to rewrite the argument of the exponential, $-\frac{1}{2} \mathbf{x}^\top A \mathbf{x} - \mathbf{x}^\top \mathbf{y}$, into the form $-\frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top A (\mathbf{x} - \mathbf{x}_0) + C$, where $\mathbf{x}_0$ and C are constants with respect to $\mathbf{x}$.

Expanding this target form, we get:

$$\begin{aligned} -\frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top A (\mathbf{x} - \mathbf{x}_0) &= -\frac{1}{2} (\mathbf{x}^\top - \mathbf{x}_0^\top) A (\mathbf{x} - \mathbf{x}_0) \\ &= -\frac{1}{2} (\mathbf{x}^\top A \mathbf{x} - \mathbf{x}^\top A \mathbf{x}_0 - \mathbf{x}_0^\top A \mathbf{x} + \mathbf{x}_0^\top A \mathbf{x}_0) \end{aligned} \quad (\text{A.2})$$

Since A is symmetric ($A = A^\top$), the scalar term $\mathbf{x}_0^\top A \mathbf{x}$ is equal to its own transpose: $(\mathbf{x}_0^\top A \mathbf{x})^\top = \mathbf{x}^\top A^\top \mathbf{x}_0 = \mathbf{x}^\top A \mathbf{x}_0$. Thus, the two cross-terms are equal.

$$-\frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top A (\mathbf{x} - \mathbf{x}_0) = -\frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{x}^\top A \mathbf{x}_0 - \frac{1}{2} \mathbf{x}_0^\top A \mathbf{x}_0 \quad (\text{A.3})$$

Comparing this to the original exponent, $-\frac{1}{2} \mathbf{x}^\top A \mathbf{x} - \mathbf{x}^\top \mathbf{y}$, we can equate the terms linear in $\mathbf{x}$:

$$-\mathbf{x}^\top \mathbf{y} = \mathbf{x}^\top A \mathbf{x}_0 \implies A \mathbf{x}_0 = -\mathbf{y} \quad (\text{A.4})$$

Since A is positive-definite, it is invertible. We can solve for x_0 :

$$x_0 = -A^{-1}y \quad (\text{A.5})$$

With this definition of x_0 , the original exponent can be written as:

$$-\frac{1}{2}x^T A x - x^T y = -\frac{1}{2}(x + A^{-1}y)^T A (x + A^{-1}y) + \frac{1}{2}(A^{-1}y)^T A (A^{-1}y) \quad (\text{A.6})$$

Let's simplify the constant term (the term not involving x):

$$\begin{aligned} \frac{1}{2}(A^{-1}y)^T A (A^{-1}y) &= \frac{1}{2}y^T (A^{-1})^T A A^{-1}y \\ &= \frac{1}{2}y^T A^{-1} A A^{-1}y \quad (\text{since } (A^{-1})^T = (A^T)^{-1} = A^{-1}) \\ &= \frac{1}{2}y^T I A^{-1}y = \frac{1}{2}y^T A^{-1}y \end{aligned} \quad (\text{A.7})$$

So, the exponent is:

$$-\frac{1}{2}x^T A x - x^T y = -\frac{1}{2}(x + A^{-1}y)^T A (x + A^{-1}y) + \frac{1}{2}y^T A^{-1}y \quad (\text{A.8})$$

Step 2: Change of Variables (Translation)

Substituting the completed square back into the integral:

$$I = \int \prod_{n=1}^N dx_n \exp \left[-\frac{1}{2}(x + A^{-1}y)^T A (x + A^{-1}y) + \frac{1}{2}y^T A^{-1}y \right] \quad (\text{A.9})$$

The term $\exp(\frac{1}{2}y^T A^{-1}y)$ is constant with respect to x and can be factored out of the integral:

$$I = \exp \left(\frac{1}{2}y^T A^{-1}y \right) \int \prod_{n=1}^N dx_n \exp \left[-\frac{1}{2}(x + A^{-1}y)^T A (x + A^{-1}y) \right] \quad (\text{A.10})$$

Now, we perform a change of variables. Let $z = x + A^{-1}y$. This is a simple translation of the coordinate system. The differential element $\prod dx_n$ transforms as $\prod dz_n$, as the Jacobian of this transformation is 1. The limits of integration remain from $-\infty$ to $+\infty$. The integral becomes:

$$I = \exp \left(\frac{1}{2}y^T A^{-1}y \right) \int \prod_{n=1}^N dz_n \exp \left(-\frac{1}{2}z^T A z \right) \quad (\text{A.11})$$

The problem is now reduced to evaluating the simpler, centered Gaussian integral:

$$I_0 = \int \prod dz_n \exp \left(-\frac{1}{2}z^T A z \right) \quad (\text{A.12})$$

Step 3: Diagonalization

To compute I_0 , we diagonalize the matrix A . Since A is a real symmetric matrix, it is orthogonally diagonalizable:

$$A = PDP^T \quad (\text{A.13})$$

where P is an orthogonal matrix ($PP^T = P^TP = I$) whose columns are the orthonormal eigenvectors of A , and D is a diagonal matrix whose entries are the corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_N$. Substituting this into the quadratic form $z^T A z$:

$$z^T A z = z^T (PDP^T) z = (z^T P) D (P^T z) = (P^T z)^T D (P^T z) \quad (\text{A.14})$$

We perform another change of variables. Let $w = P^T z$. This transformation corresponds to a rotation of the coordinate system. The Jacobian determinant is $|\det(P^T)| = 1$, so the volume element is unchanged: $\prod dz_n = \prod dw_n$. The quadratic form simplifies to:

$$w^T D w = \sum_{i=1}^N \lambda_i w_i^2 \quad (\text{A.15})$$

This is because D is a diagonal matrix.

Step 4: Computing the Decoupled Integral

The integral I_0 now becomes:

$$I_0 = \int \prod_{n=1}^N dw_n \exp\left(-\frac{1}{2} \sum_{i=1}^N \lambda_i w_i^2\right) \quad (\text{A.16})$$

The exponential of a sum is the product of exponentials, which allows us to separate the multi-dimensional integral into a product of N one-dimensional integrals:

$$I_0 = \int \prod_{n=1}^N dw_n \prod_{i=1}^N \exp\left(-\frac{1}{2} \lambda_i w_i^2\right) = \prod_{i=1}^N \left(\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} \lambda_i w_i^2\right) dw_i \right) \quad (\text{A.17})$$

We use the standard formula for a 1D Gaussian integral: $\int_{-\infty}^{\infty} \exp(-au^2) du = \sqrt{\pi/a}$. In our case, $a = \lambda_i/2$.

$$\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} \lambda_i w_i^2\right) dw_i = \sqrt{\frac{\pi}{\lambda_i/2}} = \sqrt{\frac{2\pi}{\lambda_i}} \quad (\text{A.18})$$

Multiplying these N results together:

$$I_0 = \prod_{i=1}^N \sqrt{\frac{2\pi}{\lambda_i}} = (2\pi)^{\frac{N}{2}} \prod_{i=1}^N (\lambda_i)^{-\frac{1}{2}} = (2\pi)^{\frac{N}{2}} \left(\prod_{i=1}^N \lambda_i \right)^{-\frac{1}{2}} \quad (\text{A.19})$$

The determinant of a matrix is equal to the product of its eigenvalues. Thus, $\det A = \det D = \prod_{i=1}^N \lambda_i$.

$$I_0 = (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} \quad (\text{A.20})$$

Step 5: Combining the Results

Finally, we substitute the value of I_0 back into our expression for I from Step 2:

$$I = \exp\left(\frac{1}{2}\mathbf{y}^\top A^{-1}\mathbf{y}\right) \cdot I_0 = \exp\left(\frac{1}{2}\mathbf{y}^\top A^{-1}\mathbf{y}\right) (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} \quad (\text{A.21})$$

Rearranging the terms yields the final result:

$$I = (2\pi)^{\frac{N}{2}} (\det A)^{-\frac{1}{2}} \exp\left(\frac{1}{2}\mathbf{y}^\top A^{-1}\mathbf{y}\right) \quad (\text{A.22})$$

This completes the proof.

Appendix B

Calculation of the Determinant and Inverse of the a Special Matrix

Let the given $n \times n$ matrix be denoted by A_n .

$$A_n = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}_{n \times n} \quad (\text{B.1})$$

B.1 Calculation of $\det(A)$

Let D_n denote the determinant of an $n \times n$ matrix of the form described in the problem. The matrix A given in the problem has dimensions $(N - 1) \times (N - 1)$. For the sake of recurrence derivation, we analyze the matrix A_n of size $n \times n$:

$$A_n = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}_{n \times n} \quad (\text{B.2})$$

Step 1: Find a recurrence relation

We compute the determinant using cofactor expansion along the first row. Let $D_n = \det(A_n)$.

$$D_n = 2 \cdot \det(M_{1,1}) - (-1) \cdot \det(M_{1,2}) \quad (\text{B.3})$$

Here, $M_{i,j}$ represents the minor matrix obtained by removing row i and column j .

1. Removing row 1 and column 1 leaves a matrix of the same structure with size $(n - 1) \times (n - 1)$, so $\det(M_{1,1}) = D_{n-1}$.
2. Removing row 1 and column 2 leaves a matrix where the first column contains only -1 in the top position and zeros elsewhere. Expanding along this new first column yields -1

multiplied by the determinant of the remaining $(n-2) \times (n-2)$ matrix (which again has the original structure). Thus, $\det(M_{1,2}) = -1 \cdot D_{n-2}$.

Substituting these back into the expansion equation:

$$\begin{aligned} D_n &= 2(D_{n-1}) - (-1)(-1 \cdot D_{n-2}) \\ D_n &= 2D_{n-1} - D_{n-2} \end{aligned} \tag{B.4}$$

Step 2: Calculate base cases

$$\begin{aligned} \text{For } n = 1 : \quad A_1 &= [2] \implies D_1 = 2 \\ \text{For } n = 2 : \quad A_2 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \implies D_2 = (2)(2) - (-1)(-1) = 3 \end{aligned} \tag{B.5}$$

Step 3: Solve the recurrence

We observe the pattern $D_1 = 2$, $D_2 = 3$, $D_3 = 4$. We hypothesize that $D_n = n + 1$. We prove this by induction.

1. The hypothesis holds for base cases $n = 1$ and $n = 2$.
2. Assume $D_k = k + 1$ for all $k < n$.
3. Apply the recurrence:

$$\begin{aligned} D_n &= 2D_{n-1} - D_{n-2} \\ &= 2((n-1) + 1) - ((n-2) + 1) \\ &= 2n - (n-1) \\ &= 2n - n + 1 \\ &= n + 1 \end{aligned} \tag{B.6}$$

Step 4: Final Calculation

The matrix A has size $(N-1) \times (N-1)$. Therefore, we substitute $n = N-1$ into the formula:

$$\det(A) = D_{N-1} = (N-1) + 1 = N \tag{B.7}$$

B.2 Calculation of A^{-1}

Let $B = A^{-1}$. The elements of the inverse matrix are denoted as $b_{i,j}$. From the definition $AB = I$, for a fixed column j of the inverse matrix, the elements $x_i = b_{i,j}$ satisfy the linear system:

$$Ax = e_j \quad (\text{B.8})$$

where e_j is the standard basis vector. The indices i and j run from 1 to $N - 1$.

Step 1: Set up the difference equations

The structure of A corresponds to a discrete second derivative. The equation for the i -th row is:

$$-x_{i-1} + 2x_i - x_{i+1} = \delta_{i,j} \quad (\text{B.9})$$

The boundary conditions correspond to the nodes outside the matrix:

$$x_0 = 0 \quad \text{and} \quad x_N = 0 \quad (\text{B.10})$$

Step 2: Solve the homogeneous regions

For $i \neq j$, the equation is $-x_{i-1} + 2x_i - x_{i+1} = 0$, which implies $x_{i+1} - x_i = x_i - x_{i-1}$. The solution is linear in i .

1. **Region 1** ($0 \leq i \leq j$): Applying $x_0 = 0$, the solution is $x_i = c_1 i$.
2. **Region 2** ($j \leq i \leq N$): Applying $x_N = 0$, the solution is $x_i = c_2(N - i)$.

Step 3: Match continuity at $i = j$

The solution must be continuous at $i = j$:

$$c_1 j = c_2(N - j) \implies c_2 = c_1 \frac{j}{N - j} \quad (\text{B.11})$$

Step 4: Apply the jump condition at $i = j$

Substitute the neighbors into the non-homogeneous equation at $i = j$:

$$-x_{j-1} + 2x_j - x_{j+1} = 1 \quad (\text{B.12})$$

Using the linear forms derived above (x_{j-1} from Region 1, x_{j+1} from Region 2):

$$\begin{aligned} -[c_1(j - 1)] + 2[c_1 j] - [c_2(N - j - 1)] &= 1 \\ c_1(j + 1) - c_2(N - j - 1) &= 1 \end{aligned} \quad (\text{B.13})$$

Substitute $c_2 = c_1 \frac{j}{N - j}$:

$$\begin{aligned}
c_1(j+1) - \left(c_1 \frac{j}{N-j}\right)(N-j-1) &= 1 \\
c_1 \left[(j+1) - \frac{j(N-j-1)}{N-j} \right] &= 1
\end{aligned} \tag{B.14}$$

Multiplying by $(N-j)$ to clear the denominator:

$$\begin{aligned}
c_1 [(j+1)(N-j) - j(N-j-1)] &= N-j \\
c_1 [(jN - j^2 + N - j) - (jN - j^2 - j)] &= N-j \\
c_1 [N] &= N-j
\end{aligned} \tag{B.15}$$

Thus, we find the coefficients:

$$c_1 = \frac{N-j}{N}, \quad c_2 = \frac{j}{N} \tag{B.16}$$

Step 5: Final Matrix Elements

Substituting the constants back into the piecewise expressions: For $i \leq j$: $x_i = \frac{N-j}{N}i$. For $i \geq j$: $x_i = \frac{j}{N}(N-i)$.

The elements of the inverse matrix are given by:

$$(A^{-1})_{i,j} = \frac{1}{N} \min(i, j)(N - \max(i, j)) \tag{B.17}$$

Appendix C

Evaluation of the Fresnel Integral

The value of the **Fresnel Integral** is:

$$\int_{-\infty}^{\infty} e^{ix^2} dx = \sqrt{\frac{\pi}{2}}(1 + i) = \sqrt{\pi}e^{i\pi/4} \quad (\text{C.1})$$

C.1 Derivation (Using Contour Integration)

Step 1: Define the Contour

We consider the complex function $f(z) = e^{iz^2}$, where z is a complex variable. We construct a closed path (contour) C in the complex plane. This path is a sector of a circle, composed of three parts:

1. **Path C_1 :** A line segment along the real axis from 0 to R .
2. **Path C_2 :** A circular arc of radius R , centered at the origin, running counter-clockwise from R to $Re^{i\pi/4}$.
3. **Path C_3 :** A line segment from $Re^{i\pi/4}$ back to the origin 0.

We will eventually let $R \rightarrow \infty$.

Step 2: Apply Cauchy's Integral Theorem

The function $f(z) = e^{iz^2}$ is analytic over the entire complex plane (it is an entire function) as it has no singularities. According to **Cauchy's Integral Theorem**, its integral over any closed path C is zero:

$$\oint_C e^{iz^2} dz = 0 \quad (\text{C.2})$$

This closed-loop integral can be split into the sum of integrals over the three paths:

$$\int_{C_1} e^{iz^2} dz + \int_{C_2} e^{iz^2} dz + \int_{C_3} e^{iz^2} dz = 0 \quad (\text{C.3})$$

Step 3: Evaluate the Integral on Each Path

1. Integral along Path C_1 (The part we want to find)

On path C_1 , we have $z = x$ (a real number) and $dz = dx$. Therefore:

$$\lim_{R \rightarrow \infty} \int_{C_1} e^{iz^2} dz = \int_0^{\infty} e^{ix^2} dx \quad (\text{C.4})$$

This is exactly half of the integral we wish to compute, since the integrand e^{ix^2} is an even function.

2. Integral along Path C_2 (Show it vanishes as $R \rightarrow \infty$)

On path C_2 , we parameterize $z = Re^{i\theta}$, where θ varies from 0 to $\pi/4$. Then $dz = iRe^{i\theta}d\theta$ and $z^2 = R^2e^{i2\theta}$. The integral becomes:

$$\begin{aligned} \int_{C_2} e^{iz^2} dz &= \int_0^{\pi/4} \exp(i(R^2e^{i2\theta})) \cdot iRe^{i\theta} d\theta \\ &= \int_0^{\pi/4} \exp(iR^2(\cos(2\theta) + i\sin(2\theta))) \cdot iRe^{i\theta} d\theta \\ &= \int_0^{\pi/4} \exp(-R^2\sin(2\theta)) \exp(i(R^2\cos(2\theta) + \theta)) \cdot iRd\theta \end{aligned} \quad (C.5)$$

Taking the magnitude:

$$\left| \int_{C_2} e^{iz^2} dz \right| \leq \int_0^{\pi/4} |\exp(-R^2\sin(2\theta))| \cdot Rd\theta = \int_0^{\pi/4} R \exp(-R^2\sin(2\theta)) d\theta \quad (C.6)$$

On the interval $[0, \pi/4]$, 2θ is in $[0, \pi/2]$. We can use Jordan's inequality, which states $\sin(x) \geq \frac{2x}{\pi}$ for $x \in [0, \pi/2]$. Thus, $\sin(2\theta) \geq \frac{4\theta}{\pi}$.

$$\begin{aligned} \left| \int_{C_2} e^{iz^2} dz \right| &\leq \int_0^{\pi/4} R \exp(-R^2(4\theta/\pi)) d\theta \\ &= R \left[\frac{-\pi}{4R^2} \exp(-4R^2\theta/\pi) \right]_0^{\pi/4} \\ &= \frac{\pi}{4R} (1 - \exp(-R^2)) \end{aligned} \quad (C.7)$$

As $R \rightarrow \infty$, this expression approaches 0. Therefore:

$$\lim_{R \rightarrow \infty} \int_{C_2} e^{iz^2} dz = 0 \quad (C.8)$$

3. Integral along Path C_3 (Connection to the Gaussian Integral)

On path C_3 , we parameterize $z = re^{i\pi/4}$, where r varies from R to 0. Then $dz = e^{i\pi/4}dr$ and $z^2 = (re^{i\pi/4})^2 = r^2e^{i\pi/2} = r^2i$. The integral becomes:

$$\int_{C_3} e^{iz^2} dz = \int_R^0 \exp(i(r^2i)) e^{i\pi/4} dr = \int_R^0 \exp(-r^2) e^{i\pi/4} dr \quad (C.9)$$

Reversing the limits of integration and factoring out the constant:

$$= -e^{i\pi/4} \int_0^R \exp(-r^2) dr \quad (C.10)$$

As $R \rightarrow \infty$, we get the well-known Gaussian integral:

$$\lim_{R \rightarrow \infty} \int_{C_3} e^{iz^2} dz = -e^{i\pi/4} \int_0^\infty \exp(-r^2) dr \quad (\text{C.11})$$

We know that the Gaussian integral $\int_0^\infty \exp(-x^2) dx = \frac{\sqrt{\pi}}{2}$. So:

$$\lim_{R \rightarrow \infty} \int_{C_3} e^{iz^2} dz = -e^{i\pi/4} \frac{\sqrt{\pi}}{2} \quad (\text{C.12})$$

Step 4: Combine the Results

Returning to the equation from Step 2 and taking the limit as $R \rightarrow \infty$:

$$\left(\int_0^\infty e^{ix^2} dx \right) + (0) + \left(-e^{i\pi/4} \frac{\sqrt{\pi}}{2} \right) = 0 \quad (\text{C.13})$$

Rearranging the terms, we find:

$$\int_0^\infty e^{ix^2} dx = e^{i\pi/4} \frac{\sqrt{\pi}}{2} \quad (\text{C.14})$$

Step 5: Calculate the Final Integral

The integral we want to evaluate is from $-\infty$ to ∞ . Since the integrand $e^{ix^2} = \cos(x^2) + i \sin(x^2)$ is an even function (i.e., $f(-x) = f(x)$), we have:

$$\begin{aligned} \int_{-\infty}^\infty e^{ix^2} dx &= 2 \int_0^\infty e^{ix^2} dx \\ &= 2 \cdot \left(e^{i\pi/4} \frac{\sqrt{\pi}}{2} \right) = \sqrt{\pi} e^{i\pi/4} \end{aligned} \quad (\text{C.15})$$

Finally, we can use Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, to expand $e^{i\pi/4}$:

$$e^{i\pi/4} = \cos(\pi/4) + i \sin(\pi/4) = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} = \frac{1+i}{\sqrt{2}} \quad (\text{C.16})$$

Substituting this into our result gives:

$$\int_{-\infty}^\infty e^{ix^2} dx = \sqrt{\pi} \cdot \frac{1+i}{\sqrt{2}} = \sqrt{\frac{\pi}{2}} (1+i) \quad (\text{C.17})$$

C.2 Conclusion

The result of the integral is a complex number. Its real and imaginary parts correspond to two other important integrals:

$$\begin{cases} \int_{-\infty}^\infty \cos(x^2) dx = \sqrt{\frac{\pi}{2}} \\ \int_{-\infty}^\infty \sin(x^2) dx = \sqrt{\frac{\pi}{2}} \end{cases} \quad (\text{C.18})$$